# Supplementary material

Evaluating Crowd-Label Supervision for Selective Facial-Expression Recognition

SN Computer Science

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## How to use this supplement

Sections S1–S7 retain the complete diagnostics for the original experiments. Section S8 collects the fixed model architecture and the local composition prototype. Section S9 records acquisition, annotation alignment, execution chronology and reproduction routes. The final sections report the new target, vocabulary and pipeline follow-ups. The main article groups these findings more compactly; the section numbering here identifies the underlying evidence rather than matching each main-text subsection.

## Contents

S1 Effects of supervision targets and checkpoint selection . . . 2
S2 Distributional agreement and backbone sensitivity . . . 5
S3 Accepted-image sets and annotation references . . . 8
S4 Effects of unsupported annotation categories . . . 11
S5 Sensitivity to confidence scores . . . 13
S6 Calibration-to-test transfer . . . 15
S7 External clip evaluation and diagnostic analyses . . . 16
S8 Model and prototype implementation . . . 23
S9 Execution provenance and reproduction . . . 25
S10 Final ranking and temporal-aggregation diagnostics . . . 26
S11 New target, vocabulary and stopping-rule comparisons . . . 28

## Notation and interpretation

Original numerical summaries are generated from `reported_results.csv`. The additional score-sensitivity summaries are generated from the three `selector` CSV ledgers in the consolidated Online Resource 1. S denotes Soft, U Uniform, H Hard, and T Tie-hard. Differences are percentage points. The three seeds are fixed; they do not represent all possible retraining outcomes. Training phases reuse one image source. A later analysis tests frozen models on acted videos and is reported separately. Neither analysis establishes a benefit for natural communication. Global 80% first selects images from the full population, then reports the retained images in each stratum. Within-stratum 80% instead ranks and selects images separately inside that stratum. Supported-only risk uses only votes for categories the model can predict. It is undefined when no votes fall in those categories.

# S1 Effects of supervision targets and checkpoint selection

Table S1: Original four-condition paired seed contrasts at 80% coverage

| State | Contrast | Seed 17 | Seed 42 | Seed 89 |
|---|---|---|---|---|
| selected | Soft - Uniform | -0.3473 | -0.4313 | -0.3550 |
| selected | Soft - Hard | -2.1908 | -3.7328 | -1.9084 |
| selected | Tie-hard - Hard | -0.0153 | -1.0992 | -0.2977 |
| selected | Soft - Tie-hard | -2.1756 | -2.6336 | -1.6107 |
| epoch12 | Soft - Uniform | -0.3282 | -0.3931 | -0.2290 |
| epoch12 | Soft - Hard | -1.1794 | -1.4198 | -1.4542 |
| epoch12 | Tie-hard - Hard | -0.2824 | -0.5076 | -0.4313 |
| epoch12 | Soft - Tie-hard | -0.8969 | -0.9122 | -1.0229 |

Selected epochs for all five Swin conditions are consolidated in Table S3; every fixed-epoch sensitivity checkpoint is epoch 12.

Table S2: Secondary matched Tie-Uniform comparison at 80% coverage. Risks are percentages; differences and conditional 95% intervals are percentage points

| State | Tie-Uniform | Soft | Difference | Interval |
|---|---|---|---|---|
| selected | 24.272 | 23.684 | -0.588 | [-0.830, -0.281] |
| epoch12 | 23.939 | 23.632 | -0.307 | [-0.585, -0.024] |

Table S3: Selected epochs under the common distribution-distance criterion

| Targets | Seed 17 | Seed 42 | Seed 89 |
|---|---|---|---|
| Hard | 3 | 2 | 5 |
| Tie-hard | 4 | 4 | 5 |
| Uniform | 9 | 6 | 10 |
| Tie-Uniform | 12 | 8 | 7 |
| Soft | 9 | 10 | 10 |

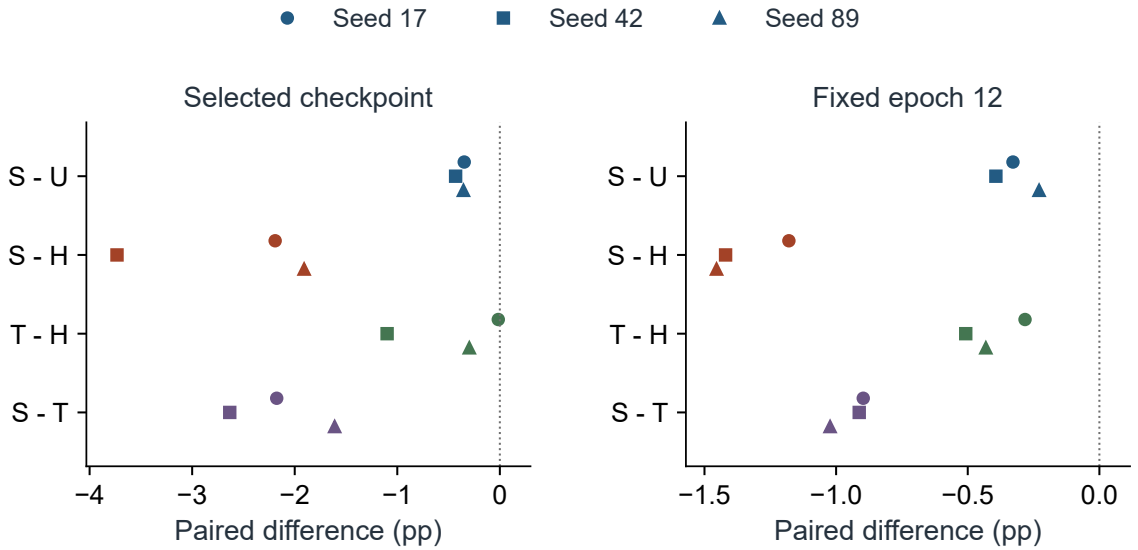



Figure S1: Original four conditions: individual paired seed differences at the common-coverage endpoint. (a) Selected and (b) fixed epoch-12 weights. Exact values appear in Table S1; within-row vertical offsets separate seeds without changing their difference values

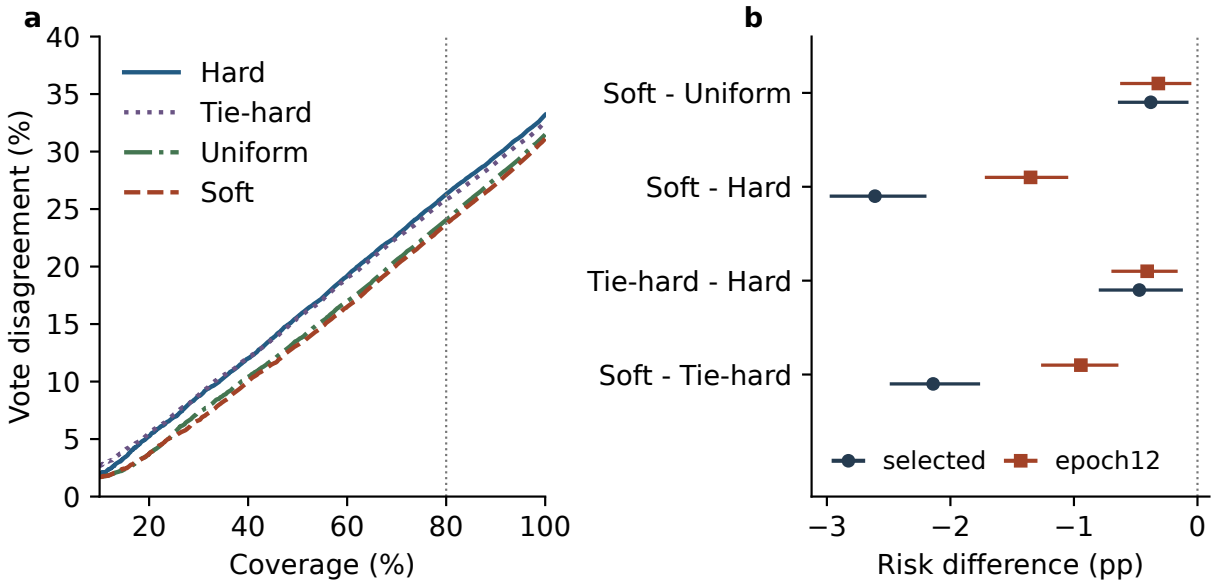



Figure S2: (a) Original four conditions: mean selected-checkpoint risk–coverage curves across three seeds; the displayed range begins at 10%, and every prefix is released. (b) Mean paired contrasts and conditional 95% pixel-cluster intervals, for selected and epoch-12 weights. Negative differences favor the first-named target. Intervals do not measure seed-population or external-domain uncertainty

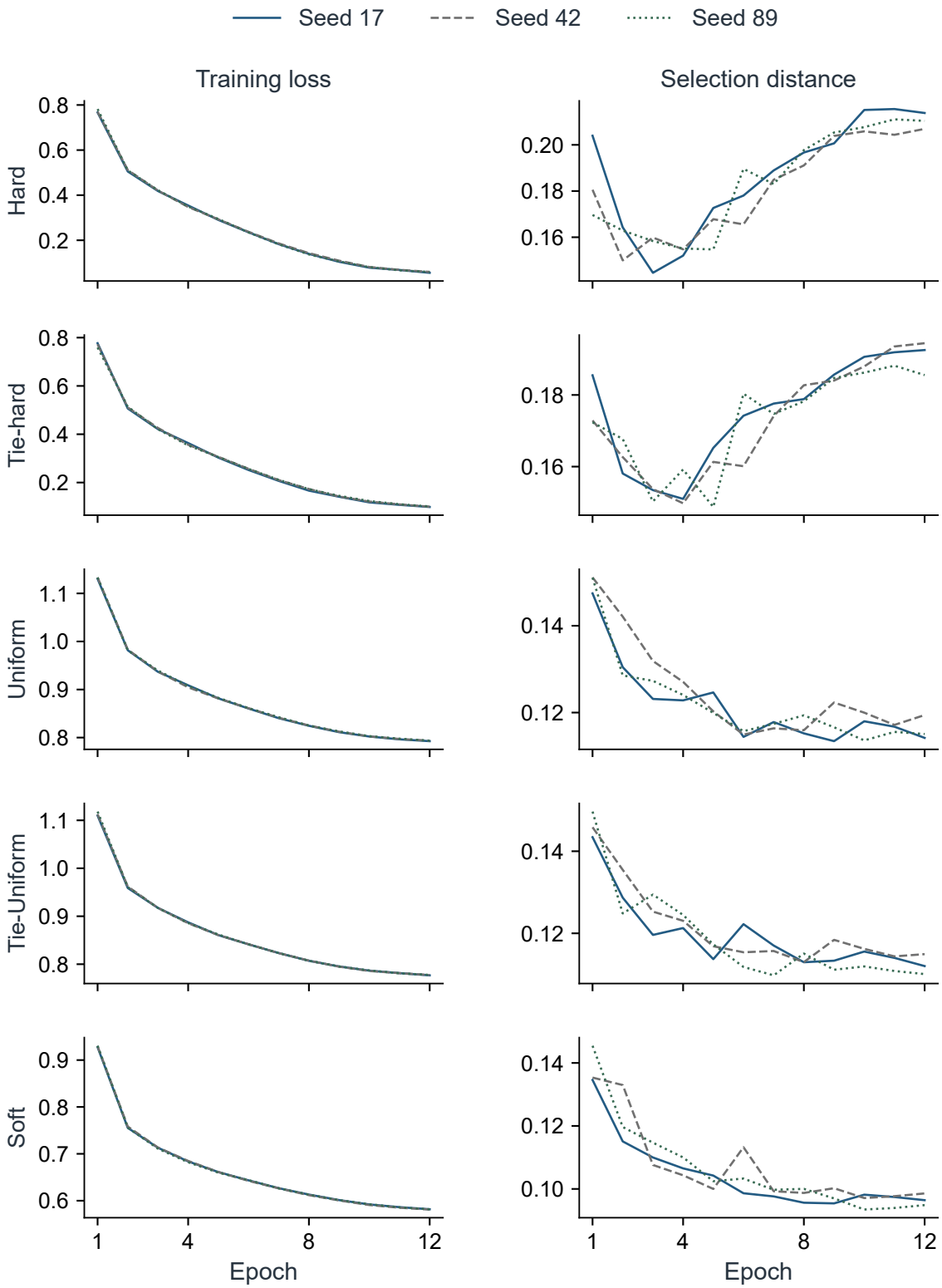



Figure S3: Training loss and selection-distance trajectories for the fifteen retained Swin runs. ResNet trajectories are supplied separately below. The target-dependent loss scales are not comparable performance endpoints.

## S2 Distributional agreement and backbone sensitivity

The distance is $D = N^{-1} \sum_i \sum_{c=1}^{10} (\tilde{p}_{ic} - q_{ic})^2$, where $\tilde{p}$ adds three zeros to the seven outputs. All states use the same $N = 3,275$ primary test rows and the original ten-category vote distributions, without renormalization. These descriptive summaries were assembled after the outcomes were known. The measure and per-state values were already recorded. No images are rejected by confidence, and no training or probability ensemble is added.

Table S4: Held-out squared distribution distance for every retained target, state and seed. Lower is better; values are distance units, not percentages

| Backbone | State | Target | Seed 17 | Seed 42 | Seed 89 | Mean |
|---|---|---|---|---|---|---|
| Swin-T | Selected | Hard | 0.153453 | 0.157580 | 0.162272 | 0.157769 |
| Swin-T | Selected | Tie-hard | 0.163318 | 0.170017 | 0.151180 | 0.161505 |
| Swin-T | Selected | Uniform | 0.115093 | 0.115385 | 0.116645 | 0.115708 |
| Swin-T | Selected | Tie-Uniform | 0.114356 | 0.116172 | 0.110653 | 0.113727 |
| Swin-T | Selected | Soft | 0.098510 | 0.097502 | 0.097491 | 0.097835 |
| Swin-T | Epoch 12 | Hard | 0.226550 | 0.225623 | 0.225190 | 0.225787 |
| Swin-T | Epoch 12 | Tie-hard | 0.209374 | 0.205987 | 0.202012 | 0.205791 |
| Swin-T | Epoch 12 | Uniform | 0.117631 | 0.117785 | 0.115561 | 0.116992 |
| Swin-T | Epoch 12 | Tie-Uniform | 0.114356 | 0.112754 | 0.111290 | 0.112800 |
| Swin-T | Epoch 12 | Soft | 0.098639 | 0.097274 | 0.097469 | 0.097794 |
| ResNet-18 | Selected | Uniform | 0.121368 | 0.123774 | 0.123949 | 0.123030 |
| ResNet-18 | Selected | Tie-Uniform | 0.119359 | 0.118507 | 0.116272 | 0.118046 |
| ResNet-18 | Selected | Soft | 0.109512 | 0.112412 | 0.109837 | 0.110587 |
| ResNet-18 | Epoch 12 | Uniform | 0.121749 | 0.122098 | 0.123936 | 0.122595 |
| ResNet-18 | Epoch 12 | Tie-Uniform | 0.115945 | 0.118419 | 0.115596 | 0.116653 |
| ResNet-18 | Epoch 12 | Soft | 0.110599 | 0.111553 | 0.110533 | 0.110895 |

The companion `distribution_distance.csv` retains full precision, prediction hashes and exact agreement with the consolidated original ledger. Soft is lower than both controls for each paired seed in both families and states. These overlapping comparisons are not independent replications, and no new confidence interval or calibration claim is attached.

The main article reports differences and conditional intervals for both checkpoint states. The following table adds absolute ResNet risks, and the figures report the retained learning trajectories and coverage comparisons. Accepted counts and fixed-image comparisons appear in Sections S4 and S3, respectively. These secondary checks were designed adaptively, reuse the same images, and do not rank architectures.

Table S5: ResNet-18 complete-vote risk (%) at 80% coverage, means across paired seeds

| Targets | Selected | Epoch 12 |
|---|---|---|
| Soft | 24.700 | 24.603 |
| Uniform | 24.527 | 24.433 |
| Tie-Uniform | 24.658 | 24.413 |

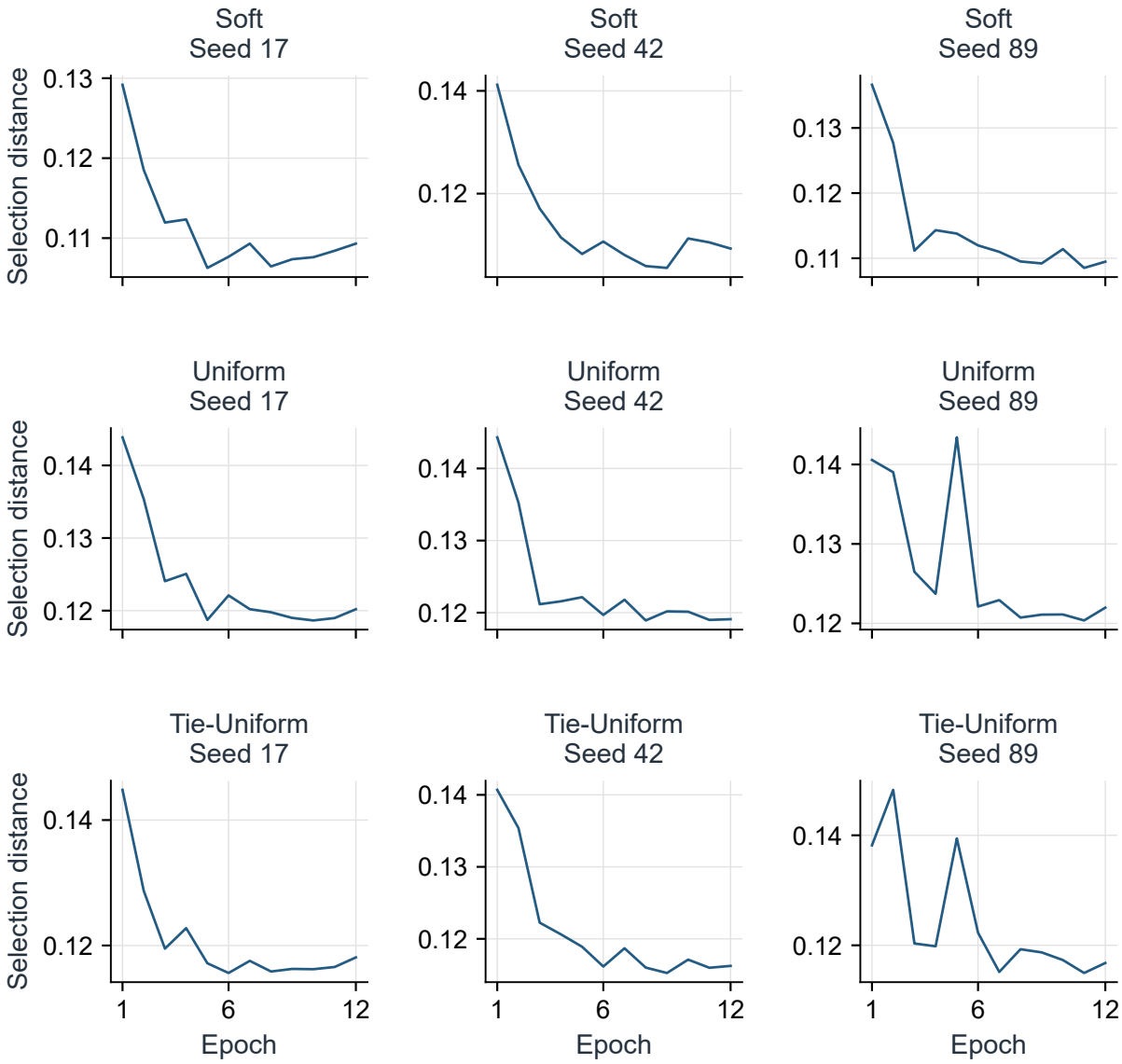



Figure S4: Selection-distance trajectories for the nine ResNet runs. Selected and epoch-12 models are both evaluated; these curves do not establish convergence

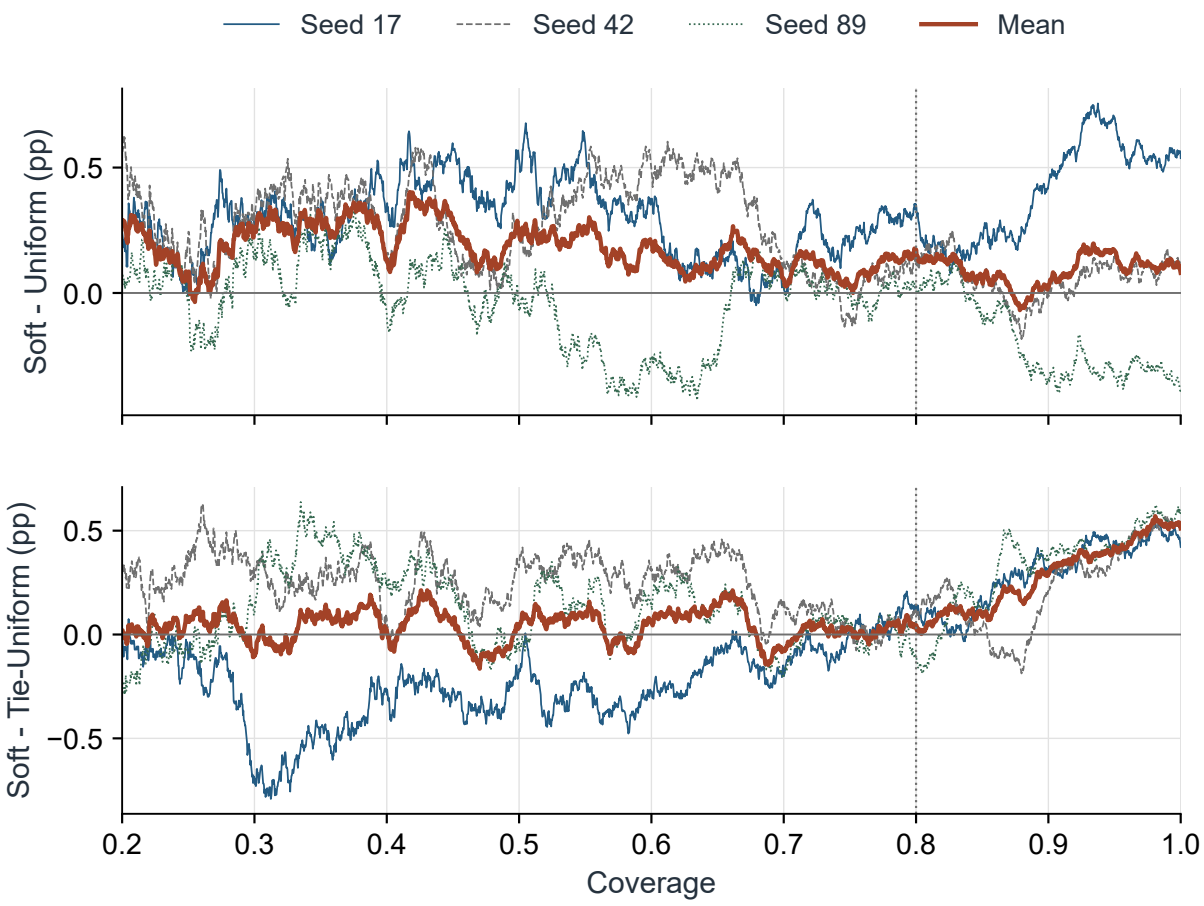



Figure S5: ResNet selected-model own-ranking risk differences at 20–100% common coverage. Thin curves are paired seeds and thick curves their mean. The dotted marker identifies the prespecified 80% secondary endpoint. These descriptive curves have no simultaneous uncertainty bands and do not compare predictions on identical accepted images

# S3 Accepted-image sets and annotation references

Table S6: Swin-T per-seed fixed-image cross-evaluation

| State | Predictor | Seed | Uniform set | Soft set |
|---|---|---|---|---|
| selected | Uniform | 17 | 24.0496 | 24.0878 |
| selected | Soft | 17 | 24.1145 | 23.7023 |
| selected | Uniform | 42 | 24.1718 | 24.1908 |
| selected | Soft | 42 | 24.2023 | 23.7405 |
| selected | Uniform | 89 | 23.9656 | 23.8321 |
| selected | Soft | 89 | 24.1260 | 23.6107 |
| epoch12 | Uniform | 17 | 23.9733 | 24.0076 |
| epoch12 | Soft | 17 | 23.9389 | 23.6450 |
| epoch12 | Uniform | 42 | 24.0840 | 23.8817 |
| epoch12 | Soft | 42 | 24.2595 | 23.6908 |
| epoch12 | Uniform | 89 | 23.7901 | 23.6489 |
| epoch12 | Soft | 89 | 23.9122 | 23.5611 |

Risk entries are complete-vote disagreement percentages at exactly 2,620 accepted images.

Table S7: Swin-T accepted-set overlap for Soft and Uniform

| State | Seed | Intersection | Union | Jaccard | Overlap fraction |
|---|---|---|---|---|---|
| selected | 17 | 2397 | 2843 | 0.8431 | 0.9149 |
| selected | 42 | 2381 | 2859 | 0.8328 | 0.9088 |
| selected | 89 | 2402 | 2838 | 0.8464 | 0.9168 |
| epoch12 | 17 | 2394 | 2846 | 0.8412 | 0.9137 |
| epoch12 | 42 | 2391 | 2849 | 0.8392 | 0.9126 |
| epoch12 | 89 | 2404 | 2836 | 0.8477 | 0.9176 |

Overlap fraction divides the intersection by 2,620, whereas Jaccard divides by the union. The two accepted sets can share most images while their differing tails change mean risk.

### Selected-checkpoint fixed-image comparisons

Table S8: ResNet-18 selected fixed-image cross-evaluation

| Control | Predictor | Selector | Seed 17 | Seed 42 | Seed 89 |
|---|---|---|---|---|---|
| Uniform | Soft | Soft | 24.8321 | 24.7824 | 24.4847 |
| Uniform | Soft | Uniform | 25.1183 | 25.0382 | 24.7595 |
| Uniform | Uniform | Soft | 25.2214 | 25.0687 | 25.0458 |
| Uniform | Uniform | Uniform | 24.4847 | 24.6450 | 24.4504 |
| Tie-Uniform | Soft | Soft | 24.8321 | 24.7824 | 24.4847 |
| Tie-Uniform | Soft | Tie-Uniform | 25.3588 | 25.3397 | 25.1145 |
| Tie-Uniform | Tie-Uniform | Soft | 24.9656 | 24.7748 | 24.4313 |
| Tie-Uniform | Tie-Uniform | Tie-Uniform | 24.7137 | 24.6679 | 24.5916 |

Entries are complete-vote disagreement percentages at 2,620 retained images.

## Fixed-epoch fixed-image comparisons

Table S9: ResNet-18 epoch 12 fixed-image cross-evaluation

| Control | Predictor | Selector | Seed 17 | Seed 42 | Seed 89 |
|---|---|---|---|---|---|
| Uniform | Soft | Soft | 24.6069 | 24.6450 | 24.5573 |
| Uniform | Soft | Uniform | 24.8702 | 24.7519 | 24.9313 |
| Uniform | Uniform | Soft | 24.9275 | 24.8053 | 24.9847 |
| Uniform | Uniform | Uniform | 24.5344 | 24.1565 | 24.6069 |
| Tie-Uniform | Soft | Soft | 24.6069 | 24.6450 | 24.5573 |
| Tie-Uniform | Soft | Tie-Uniform | 24.7176 | 24.9618 | 24.9504 |
| Tie-Uniform | Tie-Uniform | Soft | 24.6412 | 24.8779 | 24.6756 |
| Tie-Uniform | Tie-Uniform | Tie-Uniform | 24.2824 | 24.5267 | 24.4313 |

Entries are complete-vote disagreement percentages at 2,620 retained images.

Table S10: ResNet-18 reference-specific weighted F1

| State | Targets | Original/all | Original/subset | Crowd/subset |
|---|---|---|---|---|
| selected | Soft | 0.5564 | 0.6508 | 0.9053 |
| selected | Uniform | 0.5558 | 0.6531 | 0.9093 |
| selected | Tie-Uniform | 0.5588 | 0.6572 | 0.9181 |
| epoch12 | Soft | 0.5612 | 0.6561 | 0.9110 |
| epoch12 | Uniform | 0.5563 | 0.6542 | 0.9135 |
| epoch12 | Tie-Uniform | 0.5608 | 0.6597 | 0.9209 |

All images: 3,275; identical majority subset in the last two columns: 2,484. No extra relabeling is performed. Per-seed unsupported masses and overlaps are supplied in the CSV ledger.

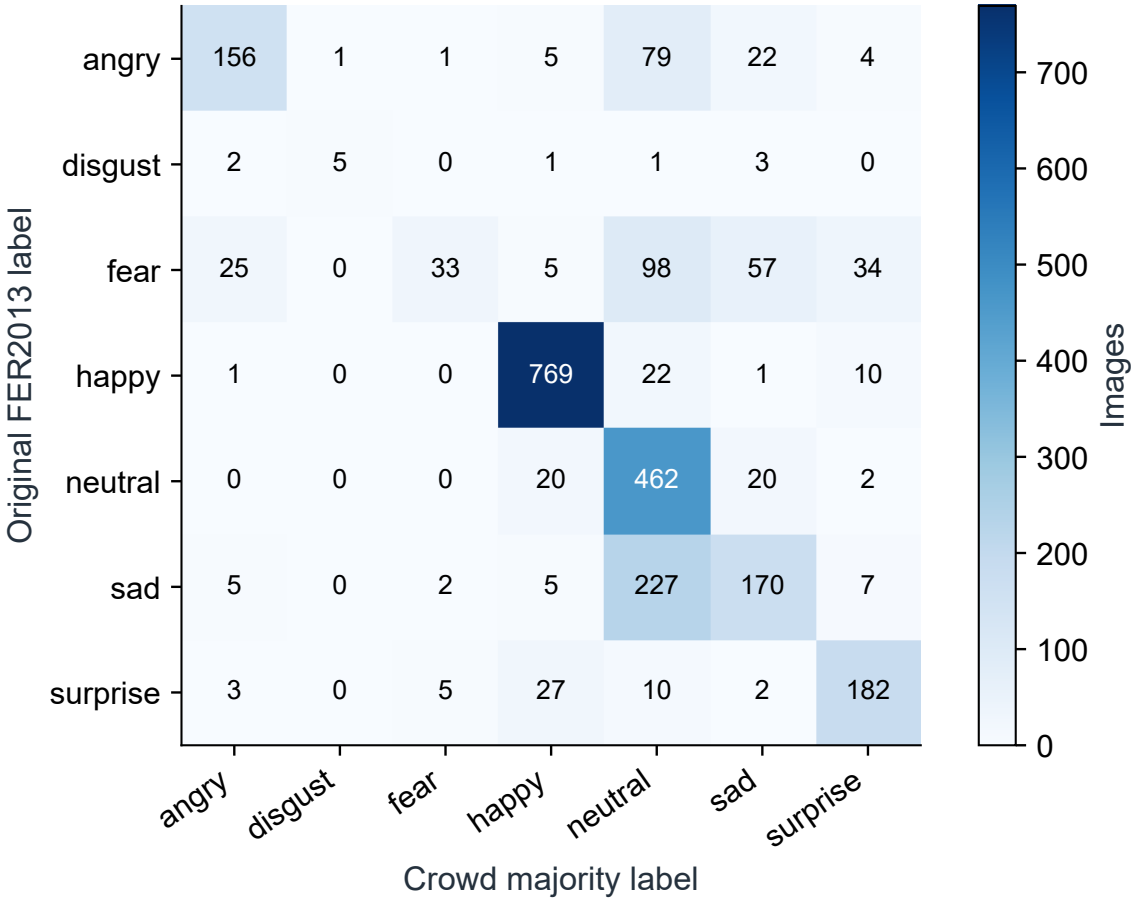



Figure S6: Original-label versus supported crowd-majority reference counts on exactly the same 2,484 primary-test images. This matrix compares references, not predictions.

# S4 Effects of unsupported annotation categories

Table S11: Original four conditions, supported-mass strata: means over three selected-checkpoint seeds. Risk columns are percentages; accepted count can vary by seed

| Targets | Stratum | Selection | N | Accepted | Risk | Supported-only |
|---|---|---|---|---|---|---|
| Hard | full | all | 1732 | 1732.00 | 19.93 | 19.93 |
| Hard | full | global80 | 1732 | 1497.33 | 14.96 | 14.96 |
| Hard | full | within80 | 1732 | 1386.00 | 12.80 | 12.80 |
| Hard | partial | all | 1527 | 1527.00 | 47.53 | 38.44 |
| Hard | partial | global80 | 1527 | 1116.33 | 41.08 | 31.30 |
| Hard | partial | within80 | 1527 | 1222.00 | 42.69 | 33.04 |
| Hard | zero | all | 16 | 16.00 | 100.00 | – |
| Hard | zero | global80 | 16 | 6.33 | 100.00 | – |
| Hard | zero | within80 | 16 | 13.00 | 100.00 | – |
| Tie-hard | full | all | 1732 | 1732.00 | 19.42 | 19.42 |
| Tie-hard | full | global80 | 1732 | 1507.00 | 14.51 | 14.51 |
| Tie-hard | full | within80 | 1732 | 1386.00 | 12.33 | 12.33 |
| Tie-hard | partial | all | 1527 | 1527.00 | 46.84 | 37.52 |
| Tie-hard | partial | global80 | 1527 | 1105.67 | 40.76 | 30.74 |
| Tie-hard | partial | within80 | 1527 | 1222.00 | 42.23 | 32.38 |
| Tie-hard | zero | all | 16 | 16.00 | 100.00 | – |
| Tie-hard | zero | global80 | 16 | 7.33 | 100.00 | – |
| Tie-hard | zero | within80 | 16 | 13.00 | 100.00 | – |
| Uniform | full | all | 1732 | 1732.00 | 18.32 | 18.32 |
| Uniform | full | global80 | 1732 | 1523.00 | 13.45 | 13.45 |
| Uniform | full | within80 | 1732 | 1386.00 | 10.94 | 10.94 |
| Uniform | partial | all | 1527 | 1527.00 | 45.71 | 36.24 |
| Uniform | partial | global80 | 1527 | 1090.00 | 38.40 | 28.24 |
| Uniform | partial | within80 | 1527 | 1222.00 | 40.58 | 30.62 |
| Uniform | zero | all | 16 | 16.00 | 100.00 | – |
| Uniform | zero | global80 | 16 | 7.00 | 100.00 | – |
| Uniform | zero | within80 | 16 | 13.00 | 100.00 | – |
| Soft | full | all | 1732 | 1732.00 | 17.93 | 17.93 |
| Soft | full | global80 | 1732 | 1529.00 | 13.23 | 13.23 |
| Soft | full | within80 | 1732 | 1386.00 | 10.42 | 10.42 |
| Soft | partial | all | 1527 | 1527.00 | 45.36 | 35.75 |
| Soft | partial | global80 | 1527 | 1080.00 | 37.71 | 27.35 |
| Soft | partial | within80 | 1527 | 1222.00 | 40.00 | 29.77 |
| Soft | zero | all | 16 | 16.00 | 100.00 | – |
| Soft | zero | global80 | 16 | 11.00 | 100.00 | – |
| Soft | zero | within80 | 16 | 13.00 | 100.00 | – |

Full means that all votes fall in the output categories; partial means that some do and some do not; zero means that none do. Risk is necessarily one in the zero stratum because the model cannot predict any category that received a vote. Global selection may leave a stratum empty; its risk is then undefined.

Table S12: Unsupported category vote mass by population and accepted-set scope

| Targets | Population | Scope | Contempt | Unknown | Not-a-face |
|---|---|---|---|---|---|
| Hard | calibration | all | 1.797 | 3.666 | 0.262 |
| Hard | calibration | global80 | 1.525 | 3.092 | 0.185 |
| Hard | test | all | 1.554 | 5.667 | 0.656 |
| Hard | test | global80 | 1.291 | 4.966 | 0.391 |
| Tie-hard | calibration | all | 1.797 | 3.666 | 0.262 |
| Tie-hard | calibration | global80 | 1.446 | 3.064 | 0.274 |
| Tie-hard | test | all | 1.554 | 5.667 | 0.656 |
| Tie-hard | test | global80 | 1.302 | 4.931 | 0.433 |
| Uniform | calibration | all | 1.797 | 3.666 | 0.262 |
| Uniform | calibration | global80 | 1.329 | 2.877 | 0.193 |
| Uniform | test | all | 1.554 | 5.667 | 0.656 |
| Uniform | test | global80 | 1.224 | 4.765 | 0.421 |
| Tie-Uniform | calibration | all | 1.797 | 3.666 | 0.262 |
| Tie-Uniform | calibration | global80 | 1.314 | 2.943 | 0.238 |
| Tie-Uniform | test | all | 1.554 | 5.667 | 0.656 |
| Tie-Uniform | test | global80 | 1.246 | 4.821 | 0.405 |
| Soft | calibration | all | 1.797 | 3.666 | 0.262 |
| Soft | calibration | global80 | 1.291 | 3.004 | 0.241 |
| Soft | test | all | 1.554 | 5.667 | 0.656 |
| Soft | test | global80 | 1.228 | 4.749 | 0.571 |

Entries are mean vote-mass percentages across three seeds, not proportions of images with that label. All-image masses are model-independent; global80 conditions on each model's ranking. The three categories do not share one semantic interpretation.

Table S13: ResNet-18 global 80% accepted counts in full/partial/zero-supported-mass strata. Entries list seeds 17, 42, 89

| Targets | Full | Partial | Zero |
|---|---|---|---|
| Soft | 1509, 1533, 1522 | 1106, 1077, 1088 | 5, 10, 10 |
| Uniform | 1520, 1528, 1522 | 1091, 1085, 1092 | 9, 7, 6 |
| Tie-Uniform | 1523, 1521, 1511 | 1088, 1090, 1101 | 9, 9, 8 |

Each seed retains 2,620 of 3,275 test images. Zero-mass population size is 16.

# S5 Sensitivity to confidence scores

This additional analysis uses already exposed test predictions. All 24 fitted models and 48 inference states are retained. Probability-only scores never use test votes to rank images. The complete new ledgers cover 20%, 50%, 80% and 100%; the table below reports 80%, with 2,620 retained images per model. Seed variation is descriptive, not a confidence interval. No test-selected score or threshold is deployed.

Table S14: All paired seed differences for score sensitivity at 80% coverage (pp)

| Backbone | State | Score | Control | Seed 17 | Seed 42 | Seed 89 |
|---|---|---|---|---|---|---|
| Swin-T | Selected | MSP | Uniform | -0.3473 | -0.4313 | -0.3550 |
| Swin-T | Selected | MSP | Tie-Uniform | -0.3779 | -0.5840 | -0.8015 |
| Swin-T | Selected | Margin | Uniform | +0.0534 | -0.3015 | -0.2099 |
| Swin-T | Selected | Margin | Tie-Uniform | +0.1412 | -0.3092 | -0.7290 |
| Swin-T | Selected | Neg. entropy | Uniform | +0.0267 | -0.1641 | -0.1412 |
| Swin-T | Selected | Neg. entropy | Tie-Uniform | -0.1336 | -0.3321 | -0.5038 |
| Swin-T | Epoch 12 | MSP | Uniform | -0.3282 | -0.3931 | -0.2290 |
| Swin-T | Epoch 12 | MSP | Tie-Uniform | -0.4351 | -0.2214 | -0.2634 |
| Swin-T | Epoch 12 | Margin | Uniform | -0.1718 | -0.2977 | +0.1031 |
| Swin-T | Epoch 12 | Margin | Tie-Uniform | -0.1794 | -0.1145 | +0.1221 |
| Swin-T | Epoch 12 | Neg. entropy | Uniform | -0.1679 | -0.4618 | -0.2595 |
| Swin-T | Epoch 12 | Neg. entropy | Tie-Uniform | -0.3168 | -0.1679 | -0.2405 |
| ResNet-18 | Selected | MSP | Uniform | +0.3473 | +0.1374 | +0.0344 |
| ResNet-18 | Selected | MSP | Tie-Uniform | +0.1183 | +0.1145 | -0.1069 |
| ResNet-18 | Selected | Margin | Uniform | +0.6221 | +0.2519 | +0.3779 |
| ResNet-18 | Selected | Margin | Tie-Uniform | +0.2519 | +0.2099 | +0.1565 |
| ResNet-18 | Selected | Neg. entropy | Uniform | +0.6069 | +0.5840 | +0.3359 |
| ResNet-18 | Selected | Neg. entropy | Tie-Uniform | +0.4656 | +0.5229 | +0.4656 |
| ResNet-18 | Epoch 12 | MSP | Uniform | +0.0725 | +0.4885 | -0.0496 |
| ResNet-18 | Epoch 12 | MSP | Tie-Uniform | +0.3244 | +0.1183 | +0.1260 |
| ResNet-18 | Epoch 12 | Margin | Uniform | +0.3664 | +0.7863 | +0.3817 |
| ResNet-18 | Epoch 12 | Margin | Tie-Uniform | +0.6145 | +0.6260 | +0.4237 |
| ResNet-18 | Epoch 12 | Neg. entropy | Uniform | +0.2595 | +0.5076 | +0.1527 |
| ResNet-18 | Epoch 12 | Neg. entropy | Tie-Uniform | +0.4580 | +0.2595 | +0.5878 |

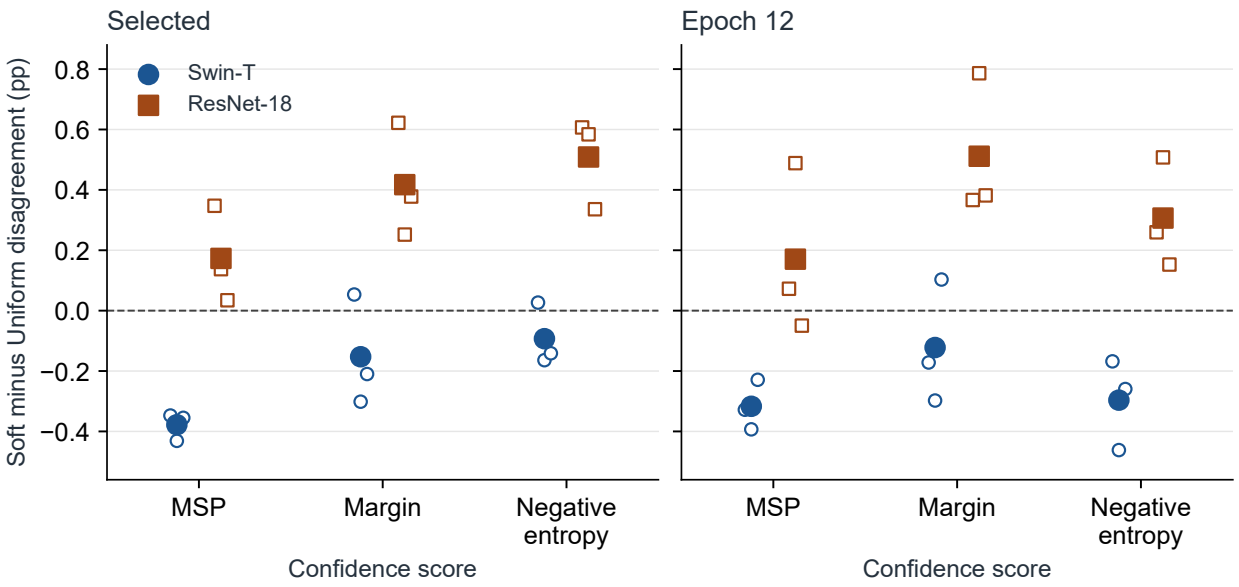



Figure S7: Probability-only score sensitivity at 80% own-ranking coverage for Soft minus Uniform. (a) Selected checkpoints; (b) fixed epoch 12. Small hollow symbols are the three paired seeds, horizontally offset only for visibility; large filled symbols are their mean. All panels share a vertical scale. Negative values favor Soft. There are no uncertainty bars; seed dispersion is not population uncertainty. The categorical score choices are not an ordered tuning path. Some Swin seed differences cross zero; no score was selected after observing these outcomes

# S6 Calibration-to-test transfer

Table S15: Original four conditions, calibration half-split holdout-minus-fit risk gaps. Means and 5th/95th percentiles are descriptive percentage points over nonempty seed–split outcomes, not confidence intervals

| Supervision | Risk target | Nonempty | Mean gap | 5th pct. | 95th pct. |
|---|---|---|---|---|---|
| Hard | 10% | 300/300 | -0.040 | -2.215 | +2.175 |
| Hard | 20% | 300/300 | +0.050 | -2.328 | +2.590 |
| Hard | 30% | 300/300 | +0.008 | -2.752 | +2.598 |
| Tie-hard | 10% | 300/300 | -0.002 | -2.169 | +2.399 |
| Tie-hard | 20% | 300/300 | +0.180 | -2.344 | +2.770 |
| Tie-hard | 30% | 300/300 | +0.072 | -2.683 | +2.859 |
| Uniform | 10% | 300/300 | +0.216 | -1.808 | +2.033 |
| Uniform | 20% | 300/300 | +0.193 | -2.021 | +2.617 |
| Uniform | 30% | 300/300 | +0.037 | -2.442 | +2.536 |
| Soft | 10% | 300/300 | +0.060 | -1.871 | +1.736 |
| Soft | 20% | 300/300 | +0.170 | -2.004 | +2.428 |
| Soft | 30% | 300/300 | +0.015 | -2.410 | +2.736 |

Each of 100 shared pixel-cluster splits uses disjoint calibration halves for fitting and checking the empirical threshold. Splits overlap one another. Minimum fitted accepted count is 100 and score ties are never split. The full ledger includes fit/holdout counts, risks, thresholds, and holdout coverage for every split, including empty policies.

Calibration: unsupported mass 5.7247%, supported ambiguity 17.9720%, annotation floor 23.6967%.

Test: unsupported mass 7.8779%, supported ambiguity 18.7878%, annotation floor 26.6656%.

These quantities use annotations only and do not depend on the fitted model. The main figure instead reports differences on the images accepted by each model's threshold. Those differences can depart from the full-population values. Neither comparison identifies a unique cause of target overshoot.

# S7 External clip evaluation and diagnostic analyses

The external unit is a video-only speech clip, not an image or frame judgment. All 1,440 clips and 24 actors are retained in the coverage denominator. The protocol, raw-vote adapter, integrity manifests, all frame/clip probabilities and five CSV ledgers are in Online Resource 8. No new training or test-fitted threshold is performed. Raw nine-category responses preserve calm and none; the merged sensitivity adds calm to neutral and leaves none unchanged. Actor intervals are in the main paper; seed dispersion below is descriptive.

Table S16: External frame eligibility by actor

| Actor | Clips | Eligible | Valid frames | Attempted frames | Max. coverage (%) |
|---|---|---|---|---|---|
| 01 | 60 | 60 | 480 | 480 | 100.00 |
| 02 | 60 | 60 | 480 | 480 | 100.00 |
| 03 | 60 | 60 | 480 | 480 | 100.00 |
| 04 | 60 | 60 | 480 | 480 | 100.00 |
| 05 | 60 | 60 | 480 | 480 | 100.00 |
| 06 | 60 | 60 | 480 | 480 | 100.00 |
| 07 | 60 | 60 | 480 | 480 | 100.00 |
| 08 | 60 | 60 | 480 | 480 | 100.00 |
| 09 | 60 | 60 | 480 | 480 | 100.00 |
| 10 | 60 | 60 | 480 | 480 | 100.00 |
| 11 | 60 | 60 | 480 | 480 | 100.00 |
| 12 | 60 | 60 | 480 | 480 | 100.00 |
| 13 | 60 | 60 | 480 | 480 | 100.00 |
| 14 | 60 | 60 | 480 | 480 | 100.00 |
| 15 | 60 | 60 | 480 | 480 | 100.00 |
| 16 | 60 | 60 | 480 | 480 | 100.00 |
| 17 | 60 | 60 | 480 | 480 | 100.00 |
| 18 | 60 | 60 | 480 | 480 | 100.00 |
| 19 | 60 | 60 | 480 | 480 | 100.00 |
| 20 | 60 | 60 | 480 | 480 | 100.00 |
| 21 | 60 | 60 | 480 | 480 | 100.00 |
| 22 | 60 | 60 | 480 | 480 | 100.00 |
| 23 | 60 | 60 | 480 | 480 | 100.00 |
| 24 | 60 | 60 | 480 | 480 | 100.00 |

Table S17: External 80% contrasts for every paired seed (pp)

| Model | State | Ref. | Score | Control | 17 | 42 | 89 |
|---|---|---|---|---|---|---|---|
| Swin-T | Sel. | Raw | MSP | Uniform | -0.052 | +2.639 | -0.755 |
| Swin-T | Sel. | Raw | MSP | Tie-Uniform | +1.007 | +2.075 | -0.252 |
| Swin-T | Sel. | Raw | Margin | Uniform | +0.773 | +3.481 | -0.920 |
| Swin-T | Sel. | Raw | Margin | Tie-Uniform | +1.120 | +2.708 | +0.174 |
| Swin-T | Sel. | Raw | Neg. entropy | Uniform | +0.217 | +1.875 | -1.024 |
| Swin-T | Sel. | Raw | Neg. entropy | Tie-Uniform | +0.755 | +1.580 | -0.625 |
| Swin-T | Sel. | Merged | MSP | Uniform | -1.276 | +2.483 | -0.903 |
| Swin-T | Sel. | Merged | MSP | Tie-Uniform | +0.747 | +1.545 | -0.226 |
| Swin-T | Sel. | Merged | Margin | Uniform | +0.009 | +3.585 | -0.859 |
| Swin-T | Sel. | Merged | Margin | Tie-Uniform | +1.259 | +2.300 | +0.339 |
| Swin-T | Sel. | Merged | Neg. entropy | Uniform | -0.825 | +1.641 | -1.146 |
| Swin-T | Sel. | Merged | Neg. entropy | Tie-Uniform | +0.807 | +1.267 | -0.625 |
| Swin-T | E12 | Raw | MSP | Uniform | -0.460 | +1.024 | +0.113 |
| Swin-T | E12 | Raw | MSP | Tie-Uniform | +0.495 | +2.535 | -0.608 |
| Swin-T | E12 | Raw | Margin | Uniform | +0.139 | +2.266 | +0.365 |
| Swin-T | E12 | Raw | Margin | Tie-Uniform | +0.130 | +3.377 | -0.703 |
| Swin-T | E12 | Raw | Neg. entropy | Uniform | -0.799 | +1.250 | -0.122 |
| Swin-T | E12 | Raw | Neg. entropy | Tie-Uniform | -0.052 | +2.405 | -0.998 |
| Swin-T | E12 | Merged | MSP | Uniform | -1.701 | -0.330 | -0.677 |
| Swin-T | E12 | Merged | MSP | Tie-Uniform | +0.226 | +1.155 | -0.434 |
| Swin-T | E12 | Merged | Margin | Uniform | -0.677 | +1.137 | -0.139 |
| Swin-T | E12 | Merged | Margin | Tie-Uniform | +0.165 | +2.326 | -0.451 |
| Swin-T | E12 | Merged | Neg. entropy | Uniform | -1.910 | -0.104 | -0.937 |
| Swin-T | E12 | Merged | Neg. entropy | Tie-Uniform | -0.078 | +1.016 | -0.825 |
| ResNet-18 | Sel. | Raw | MSP | Uniform | +1.042 | +5.521 | -1.068 |
| ResNet-18 | Sel. | Raw | MSP | Tie-Uniform | +0.955 | +4.219 | -2.917 |
| ResNet-18 | Sel. | Raw | Margin | Uniform | +1.658 | +6.345 | -0.278 |
| ResNet-18 | Sel. | Raw | Margin | Tie-Uniform | +1.806 | +4.193 | -2.118 |
| ResNet-18 | Sel. | Raw | Neg. entropy | Uniform | +0.200 | +3.377 | -1.675 |
| ResNet-18 | Sel. | Raw | Neg. entropy | Tie-Uniform | +0.486 | +2.856 | -2.387 |
| ResNet-18 | Sel. | Merged | MSP | Uniform | +1.111 | +3.776 | -0.911 |
| ResNet-18 | Sel. | Merged | MSP | Tie-Uniform | +1.189 | +3.750 | -2.717 |
| ResNet-18 | Sel. | Merged | Margin | Uniform | +1.693 | +4.332 | +0.391 |
| ResNet-18 | Sel. | Merged | Margin | Tie-Uniform | +2.161 | +3.724 | -1.389 |
| ResNet-18 | Sel. | Merged | Neg. entropy | Uniform | +0.434 | +1.510 | -1.293 |
| ResNet-18 | Sel. | Merged | Neg. entropy | Tie-Uniform | +0.816 | +2.214 | -1.866 |
| ResNet-18 | E12 | Raw | MSP | Uniform | +2.196 | +3.064 | -0.148 |
| ResNet-18 | E12 | Raw | MSP | Tie-Uniform | +1.832 | +2.474 | -1.745 |
| ResNet-18 | E12 | Raw | Margin | Uniform | +2.318 | +3.776 | +0.339 |
| ResNet-18 | E12 | Raw | Margin | Tie-Uniform | +2.691 | +3.238 | -1.094 |
| ResNet-18 | E12 | Raw | Neg. entropy | Uniform | +2.335 | +1.701 | -1.059 |
| ResNet-18 | E12 | Raw | Neg. entropy | Tie-Uniform | +2.040 | +1.866 | -1.997 |
| ResNet-18 | E12 | Merged | MSP | Uniform | +2.274 | +1.806 | -0.234 |
| ResNet-18 | E12 | Merged | MSP | Tie-Uniform | +2.144 | +2.283 | -1.771 |
| ResNet-18 | E12 | Merged | Margin | Uniform | +3.125 | +2.648 | +0.530 |
| ResNet-18 | E12 | Merged | Margin | Tie-Uniform | +3.628 | +3.116 | -0.747 |
| ResNet-18 | E12 | Merged | Neg. entropy | Uniform | +2.309 | +0.304 | -1.042 |
| ResNet-18 | E12 | Merged | Neg. entropy | Tie-Uniform | +2.075 | +1.389 | -1.753 |

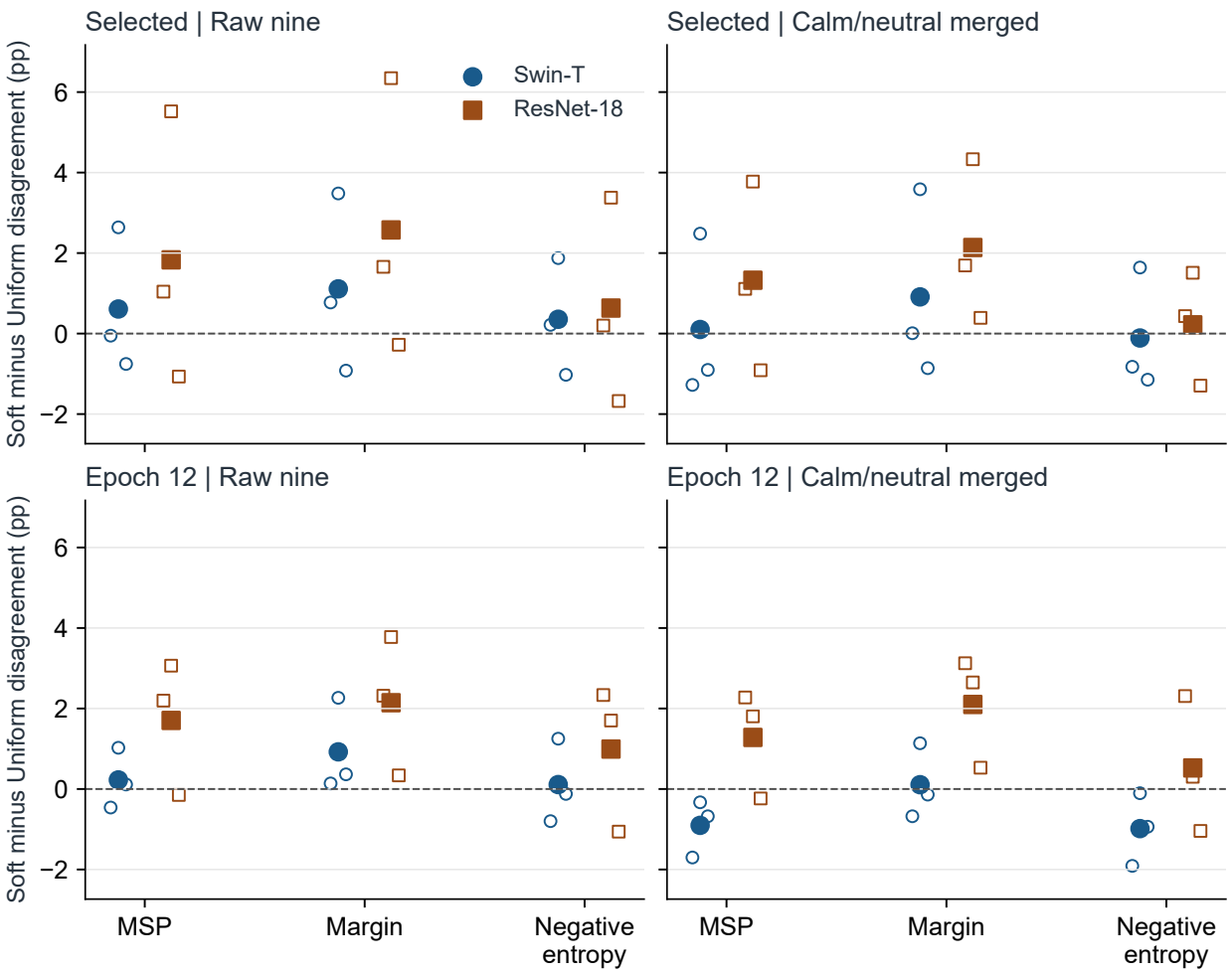



Figure S8: External confidence-score sensitivity at 80% coverage for Soft minus Uniform. Hollow marks are the three paired seeds; filled marks are their means. Panels separate checkpoint state and vote reference. All panels share the same vertical scale. These are own-ranking comparisons, not fixed-image comparisons. Seed dispersion is not an actor interval and no score is selected for deployment from this plot

Table S18: External 80% MSP fixed-image cross-evaluation: mean risk across paired seeds (%)

| Model | State | Ref. | Control | S on S | C on S | S on C | C on C |
|---|---|---|---|---|---|---|---|
| Swin-T | Sel. | Raw | Uniform | 68.37 | 68.22 | 68.07 | 67.75 |
| Swin-T | Sel. | Raw | Tie-Uniform | 68.37 | 68.00 | 67.78 | 67.42 |
| Swin-T | Sel. | Merged | Uniform | 60.72 | 60.82 | 60.77 | 60.62 |
| Swin-T | Sel. | Merged | Tie-Uniform | 60.72 | 60.39 | 60.37 | 60.03 |
| Swin-T | E12 | Raw | Uniform | 68.25 | 67.97 | 68.26 | 68.03 |
| Swin-T | E12 | Raw | Tie-Uniform | 68.25 | 67.78 | 68.04 | 67.45 |
| Swin-T | E12 | Merged | Uniform | 60.47 | 61.03 | 61.08 | 61.37 |
| Swin-T | E12 | Merged | Tie-Uniform | 60.47 | 60.34 | 60.47 | 60.16 |
| ResNet-18 | Sel. | Raw | Uniform | 71.49 | 70.47 | 70.78 | 69.66 |
| ResNet-18 | Sel. | Raw | Tie-Uniform | 71.49 | 70.87 | 71.12 | 70.73 |
| ResNet-18 | Sel. | Merged | Uniform | 63.20 | 62.54 | 62.58 | 61.88 |
| ResNet-18 | Sel. | Merged | Tie-Uniform | 63.20 | 62.52 | 62.92 | 62.46 |
| ResNet-18 | E12 | Raw | Uniform | 71.32 | 70.49 | 70.60 | 69.62 |
| ResNet-18 | E12 | Raw | Tie-Uniform | 71.32 | 70.72 | 71.05 | 70.47 |
| ResNet-18 | E12 | Merged | Uniform | 63.24 | 62.63 | 62.59 | 61.96 |
| ResNet-18 | E12 | Merged | Tie-Uniform | 63.24 | 62.60 | 62.89 | 62.35 |

S denotes Soft and C the named control. "S on C" applies Soft predictions to the clips accepted by the control. Every available cell uses the same nominal count of 1,152 clips. Per-seed values are retained in the CSV; these cross-evaluations are descriptive, not causal attribution.

Table S19: External all-eligible MSP-independent risk accounting: paired-seed means (%)

| Model | State | Ref. | Risk | Unsupported | Ambiguity | Excess | Coverage |
|---|---|---|---|---|---|---|---|
| Swin-T S | Sel. | Raw | 71.06 | 11.90 | 16.81 | 42.36 | 100.00 |
| Swin-T U | Sel. | Raw | 70.81 | 11.90 | 16.81 | 42.11 | 100.00 |
| Swin-T TU | Sel. | Raw | 70.45 | 11.90 | 16.81 | 41.75 | 100.00 |
| Swin-T S | Sel. | Merged | 64.84 | 3.14 | 20.23 | 41.47 | 100.00 |
| Swin-T U | Sel. | Merged | 64.81 | 3.14 | 20.23 | 41.44 | 100.00 |
| Swin-T TU | Sel. | Merged | 64.26 | 3.14 | 20.23 | 40.89 | 100.00 |
| Swin-T S | E12 | Raw | 70.95 | 11.90 | 16.81 | 42.25 | 100.00 |
| Swin-T U | E12 | Raw | 70.54 | 11.90 | 16.81 | 41.84 | 100.00 |
| Swin-T TU | E12 | Raw | 70.62 | 11.90 | 16.81 | 41.92 | 100.00 |
| Swin-T S | E12 | Merged | 64.59 | 3.14 | 20.23 | 41.22 | 100.00 |
| Swin-T U | E12 | Merged | 64.94 | 3.14 | 20.23 | 41.57 | 100.00 |
| Swin-T TU | E12 | Merged | 64.51 | 3.14 | 20.23 | 41.15 | 100.00 |
| ResNet-18 S | Sel. | Raw | 74.41 | 11.90 | 16.81 | 45.71 | 100.00 |
| ResNet-18 U | Sel. | Raw | 72.87 | 11.90 | 16.81 | 44.16 | 100.00 |
| ResNet-18 TU | Sel. | Raw | 73.41 | 11.90 | 16.81 | 44.70 | 100.00 |
| ResNet-18 S | Sel. | Merged | 67.62 | 3.14 | 20.23 | 44.25 | 100.00 |
| ResNet-18 U | Sel. | Merged | 66.44 | 3.14 | 20.23 | 43.07 | 100.00 |
| ResNet-18 TU | Sel. | Merged | 66.58 | 3.14 | 20.23 | 43.21 | 100.00 |
| ResNet-18 S | E12 | Raw | 74.69 | 11.90 | 16.81 | 45.99 | 100.00 |
| ResNet-18 U | E12 | Raw | 72.58 | 11.90 | 16.81 | 43.87 | 100.00 |
| ResNet-18 TU | E12 | Raw | 73.05 | 11.90 | 16.81 | 44.35 | 100.00 |
| ResNet-18 S | E12 | Merged | 67.97 | 3.14 | 20.23 | 44.60 | 100.00 |
| ResNet-18 U | E12 | Merged | 66.18 | 3.14 | 20.23 | 42.81 | 100.00 |
| ResNet-18 TU | E12 | Merged | 66.37 | 3.14 | 20.23 | 43.00 | 100.00 |

All-eligible is not relabeled 100% when a clip fails. Risk equals unsupported plus supported ambiguity plus excess before rounding. U denotes Uniform, TU Tie-Uniform. Raw/merged values score unchanged predictions; the accepted population and all-eligible coverage do not change with reference.

## Post hoc external output diagnostics

The 12 October 2026 addendum was written before these computations, but after the external outcomes were known. No fitting, peak selection or new inferential test was performed. Exports include all 36 frozen states and both populations. The displays below show all three selected Swin Soft/Uniform seeds, not only favorable ones.

Table S20: Principal selected Swin predicted-category proportions at 80% MSP (%). Columns retain every seed; each contains 1,152 clips

| | Soft | | | Uniform | | |
|---|---|---|---|---|---|---|
| Category | 17 | 42 | 89 | 17 | 42 | 89 |
| Angry | 0.87 | 1.13 | 1.22 | 1.48 | 1.74 | 1.22 |
| Disgust | 0.35 | 0.00 | 0.09 | 0.61 | 0.17 | 0.00 |
| Fear | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| Happy | 23.96 | 23.18 | 23.70 | 26.74 | 24.57 | 22.92 |
| Neutral | 68.14 | 69.97 | 67.01 | 64.58 | 62.67 | 68.32 |
| Sad | 3.04 | 2.78 | 3.65 | 3.04 | 5.56 | 3.30 |
| Surprise | 3.65 | 2.95 | 4.34 | 3.56 | 5.30 | 4.25 |

Across all 1,440 clips, raw human-vote mass (%) is angry: 14.46, disgust: 11.66, fear: 11.21, happy: 16.77, neutral: 9.06, sad: 14.21, surprise: 10.74, calm: 8.76, none: 3.14. Accepted-set vote masses are exported separately for every state and seed; model counts and reference vote mass are distinct quantities.

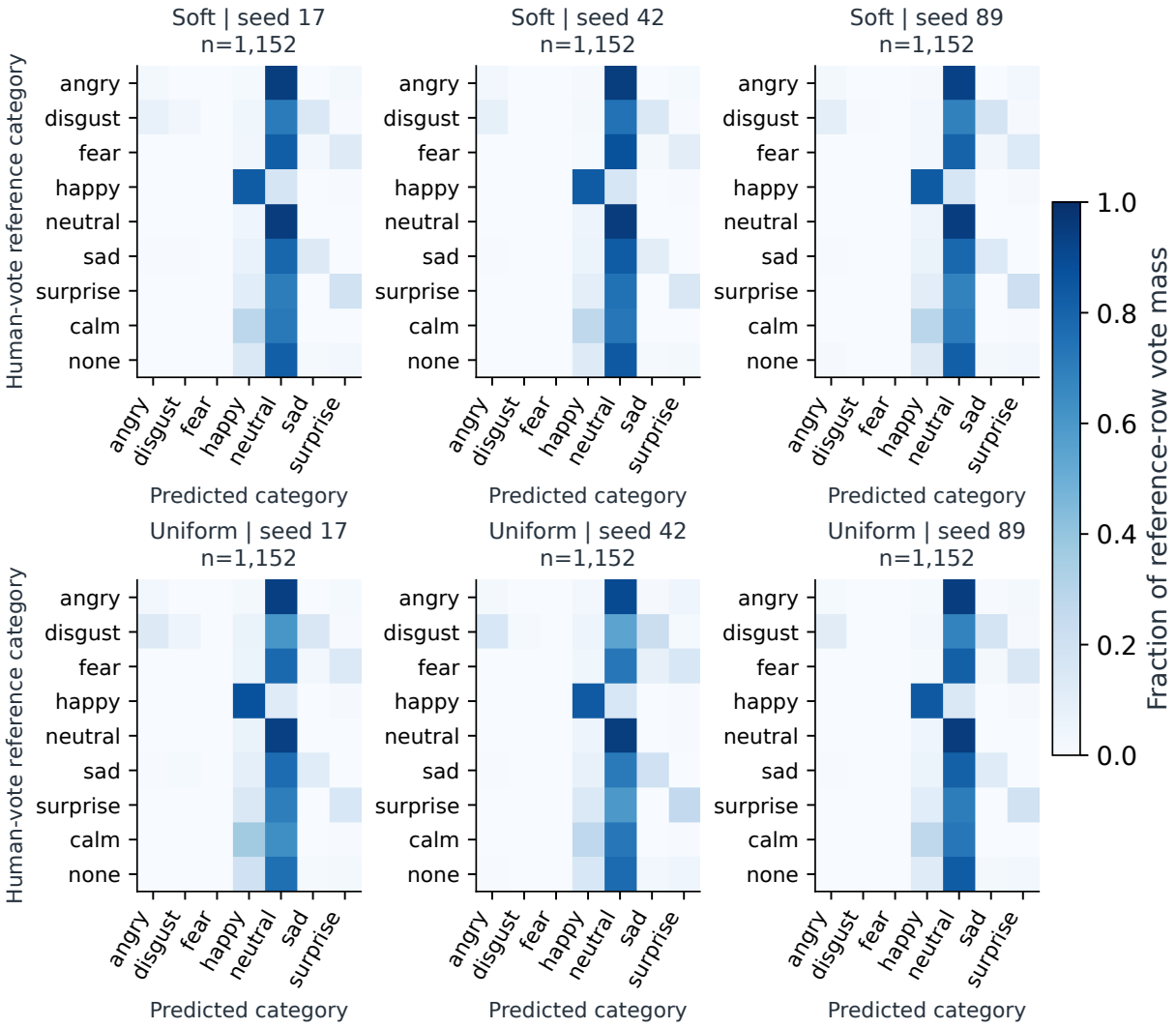



Figure S9: Human-vote reference versus predicted category for all principal selected Swin Soft/Uniform seeds at 80% MSP. Each cell divides the vote mass assigned to that predicted column by the row total; it is not single-label confusion or frame accuracy. Calm and none rows remain although the model has no corresponding outputs. Every panel contains 1,152 clips. All 36 states, both populations, raw contributions, row masses and both references are in Online Resource 8. Rows with no vote mass are undefined, not zero. The common scale is 0–1

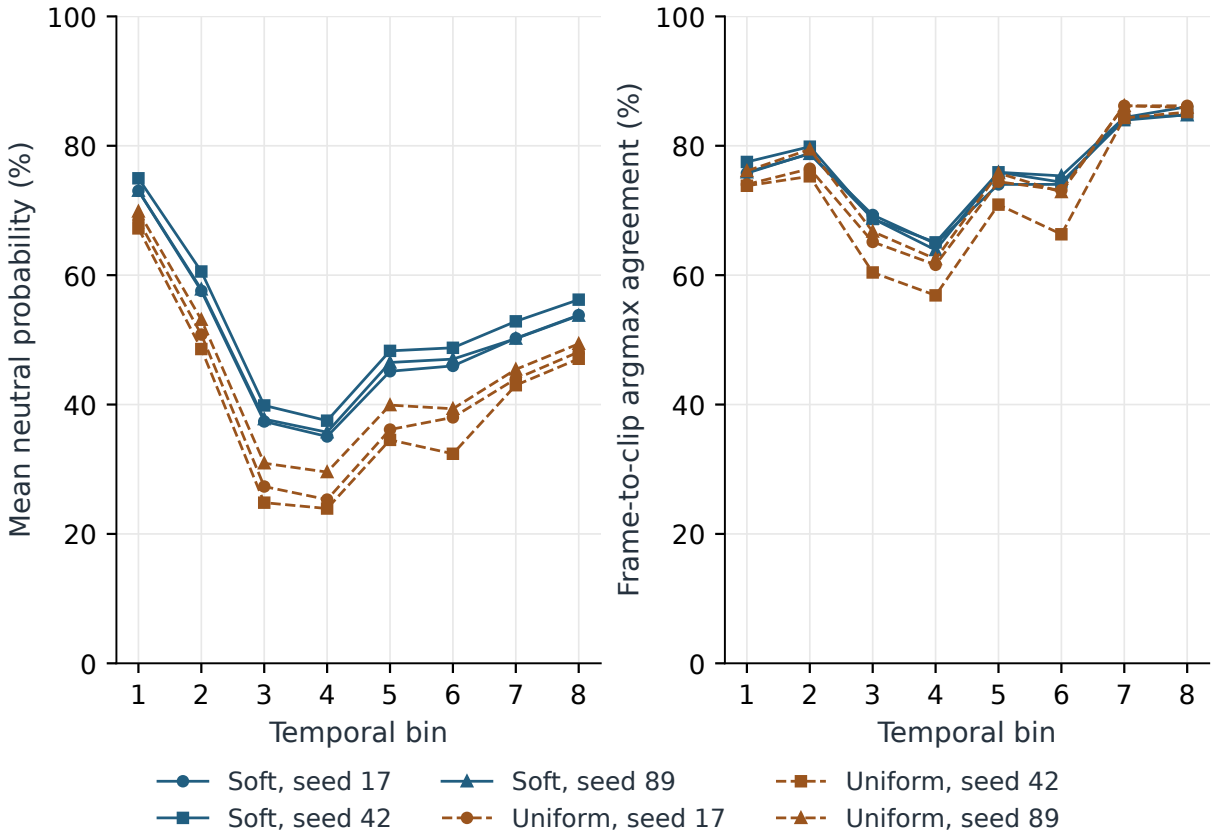



Figure S10: Predictions across all eight temporal bins for selected Swin Soft/Uniform and all three seeds. Left: mean neutral probability. Right: agreement between frame-level and mean-probability clip predictions from the same model. Each bin has 1,440 valid frames from 1,440 clips; each state has 11,520 frames. Human judgments are clip-level, so agreement is not frame accuracy. Exports include all categories and all 36 states. Temporal patterns do not establish why external disagreement occurs

Acquisition checks verify video-only speech IDs, ten responses per clip, primary response mapping, frame-bin indices, shared crop validity and canonical category order. Every clip has eight valid frames. Independently summing retained float32 frame probabilities in float64 reproduces the clip mean exactly (maximum absolute difference zero). Raw/merged scoring preserves predictions and accepted IDs. Diagnostic identities use tolerance $10^{-10}$ for vote-mass summation and $10^{-14}$ for clip aggregation; probability normalization tolerance is $10^{-6}$.

# S8 Model and prototype implementation

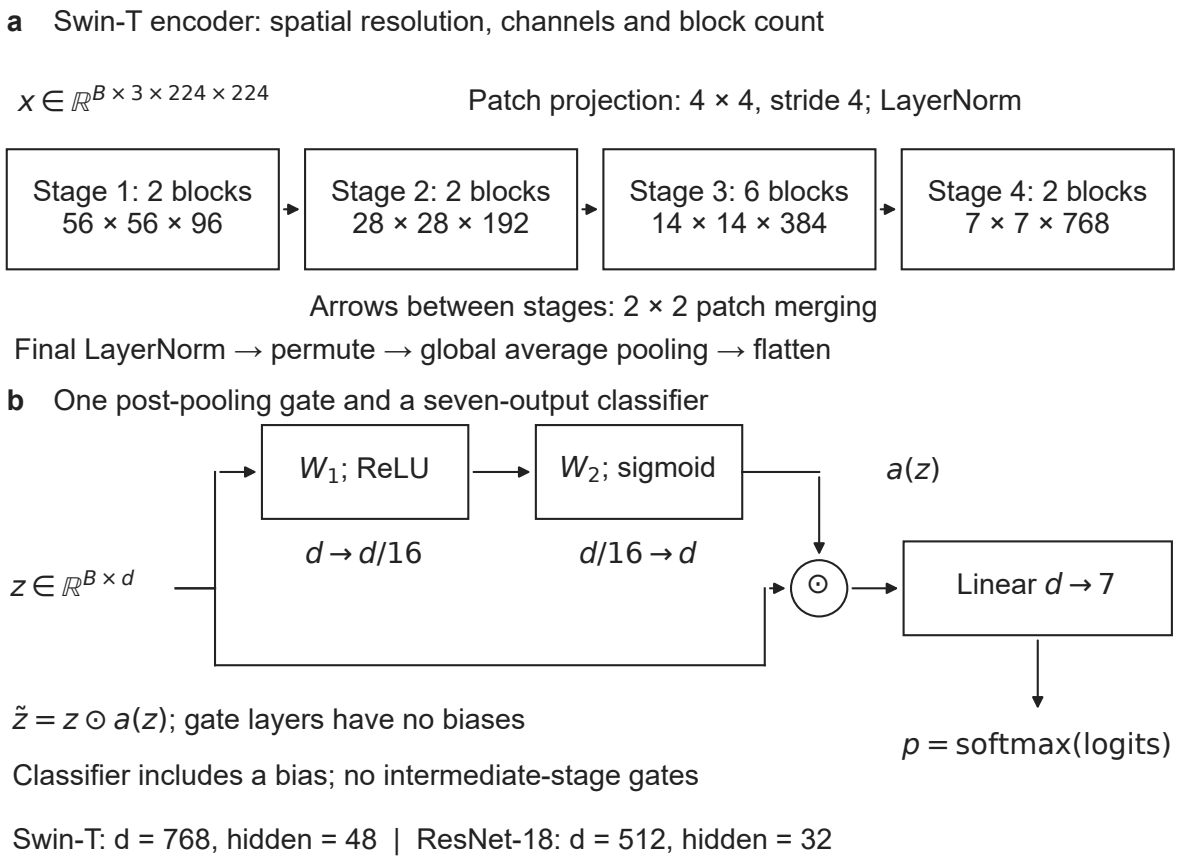



Figure S11: Original seven-output implementation. (a) Swin-T hierarchy; stage labels are $H \times W \times C$. (b) One bias-free $768 \to 48 \to 768$ ReLU/sigmoid gate acts on the pooled vector through an unchanged bypass and elementwise product, followed by a biased seven-logit classifier. The eight-expression follow-up changes this final layer to eight logits, retaining the backbone and gate. ResNet-18 uses a $512 \to 32 \to 512$ gate. No stage-wise or temporal gate is used, and architecture novelty is not claimed

Online Resource 1 supplies the implementation code; this section provides a dimensioned model diagram (Fig. S11) and a synthetic interface illustration (Fig. S12). The latter is not a participant observation or a measured prediction. The application uses MediaPipe Face Detection with model selection 1 and minimum detection confidence 0.5 (MediaPipe Face Detection; main reference list). Exactly one detected face with a valid clipped crop is eligible. The crop then uses the same grayscale, resize, normalization, class order, and model head as the experiment. Benchmark images are already cropped and do not pass through the detector. Offline predictions use the recorded BF16 evaluation path; the application uses its documented FP32 inference path, so bitwise equality across paths is not assumed.

The preview retains the original seven-output checkpoint contract. Eight-expression checkpoints belong to the research follow-up and are not deployed in the application; adding contempt to the research classifier does not silently change the interface categories or validate a new cue.

Predictions are requested no more often than once per second. A category can accompany a message only if the sender opts in, the camera is active, and a valid observation is at most 2.5 seconds old. Attachment is off by default and resets after submission. The cue is withheld when no face or multiple faces are detected, when an observation is stale, or when processing fails. These cases are not labeled neutral. The choices have not been empirically optimized. The interface labels its top-class score as uncalibrated. The confidence policies studied in the paper remain offline; they are not proven runtime protections.

Processing is local only when the Streamlit/Python backend runs on the sender's machine. The supplied configuration uses loopback, disables usage statistics and omits external ICE servers. The application processes frames in memory and does not intentionally record them. Messages stay in session memory for local preview; the application does not send them to a recipient. The category cue can itself be sensitive.

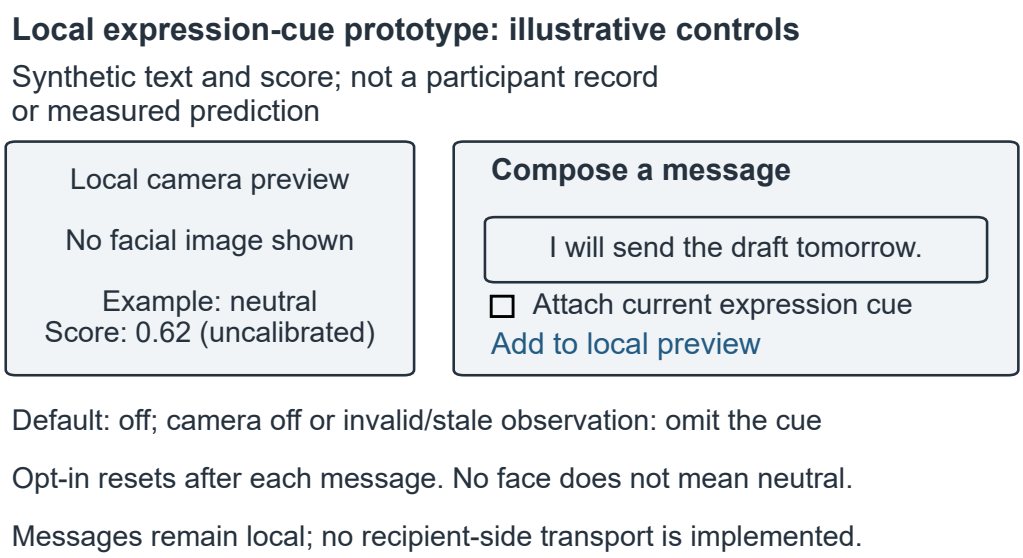



Figure S12: Synthetic illustration of the local sender-controlled interface. Text and category presentation are illustrative, not participant observations or a measured prediction. No recipient transport is implemented

These design choices are not a formal privacy guarantee and do not rule out behavior by the operating system or dependencies.

Code and numerical checks cover checkpoint metadata, gating placement, preprocessing, deterministic splits and augmentation, weighted losses, score ties, empty selections, metric definitions, and cue failure states. Runtime checks are software diagnostics, not a naturalistic usability study.

# S9 Execution provenance and reproduction

The execution records below document data handling and the timing of analysis decisions. They supplement the scientific procedures in the main Methods; they do not alter the adaptive or exploratory status of any comparison.

## FER2013 acquisition and FER+ alignment

We acquired the CSV on 20 September 2026 from Kaggle mirror `deadskull7/fer2013`, version 1, not from an official maintainer distribution. The join follows the official FER+ conversion script's row indexing, not its output filenames, which can be blank or duplicated. A separate raw-source audit checks all 35,887 original emotion IDs, pixel byte hashes and partition positions against the retained manifest. We reorder named vote columns to the fixed model class order. Online Resource 1 contains acquisition and split manifests, the original annotation records, the row adapter and the raw-source audit.

## External protocol and integrity records

We froze the local protocol on 11 October 2026 at 18:17:50 UTC, before computing external model outcomes. It records implementation, annotation, detector and checkpoint hashes. Internal outcomes and external annotation summaries were already known. The pre-outcome amendment corrects the disgust/surprise response mapping using primary-source descriptors and accuracy records; the original freeze is retained. Each RAVDESS actor archive passed the publisher's MD5 checksum, and each extracted clip retains a SHA-256 hash. Online Resource 8 includes the freeze, amendment, integrity manifests, annotation adapter and retained frame/clip predictions.

## Post hoc diagnostic addendum

A dated addendum on 12 October 2026 specified these descriptive outputs after the internal and external results were known. The addendum covers category frequencies, vote-weighted reference/prediction matrices and temporal summaries. Its status remains post hoc: no new training, fitting, frame-peak selection or confirmatory test is introduced.

## Final follow-up precision and execution repair

The 20 October follow-up retains its original scientific protocol and freeze. Its target adapter computes normalized supported proportions in float64 before float32 target and weight construction. Before permutation, comparing reconstructed seven-class Soft coefficients with the retained historical float32 raw-vote implementation identifies differences on 349 training rows, with maximum absolute difference $2.98 \times 10^{-8}$ and mean $5.29 \times 10^{-11}$ across coefficients. The annotations are unchanged. These arithmetic differences preclude claiming bitwise-identical loss coefficients across historical and new implementations; new eight-output conditions share the same adapter.

After Permutation seed 17 retained epoch 1, the next epoch failed when Windows could not allocate a worker's CPU tensor. Amendment 001 replaced spawned workers with sequential loading before new calibration/test inference. Its adapter preserves the persistent-loader sample order, stateless flips and optimizer/model RNG states, resuming after the committed epoch without repeating its updates. The source, amendment freeze and three-epoch comparison against an actual two-worker loader are retained.

Amendment 002 added one input-prefetch thread, bounded to one additional batch, and one intra-op CPU thread, matching the former workers. Sampler and transforms are unchanged. Permutation seed 17 resumed after committed epoch 2; every run records its amendment timing. A second actual two-worker comparison verifies all sample positions over three synthetic epochs. Both amendments, adapters, freezes and tests accompany the release. They preceded new calibration/test inference and changed no target, image, seed, batch size, optimizer, schedule, budget or endpoint.

An asynchronous CUDA error later interrupted selection inference in epoch 12 of eight-output Uniform, seed 89. The unchanged adapter successfully replayed that partial epoch from the verified epoch-11 model, optimizer, scheduler and RNG checkpoint. A separate recovery record reports the interruption without assigning a hardware or software cause.

`QUICKSTART.md` and the resource index identify current sources, replay commands and inference states. Compile `main.tex` and `supplement.tex`. Historical README files describe earlier editions. Fresh inference requires authorized image access; images are not redistributed. These internal checks are not external replication.

# S10 Final ranking and temporal-aggregation diagnostics

The diagnostic protocol was recorded before these additional computations, after the original outcomes were known. These are exploratory reanalyses of retained outputs, not independent validation. Source and prediction hashes, all score candidates, all-prefix calibration AURC, test risk–coverage summaries, and per-state reconciliation checks are supplied with the final evidence resource.

## S10.1 Centered logit normalization

For positive saved probabilities, centered logits are reconstructed as $z_c = \log p_c - K^{-1} \sum_j \log p_j$. The candidate score is $\max_c z_c / \|z\|_a$. Exponents are chosen independently for each fitted state on calibration AURC for complete-vote disagreement, from the ordered candidates $1, 1.5, 2, 3, 4, 6, 8, 16$. Ties retain the first candidate. All 48 calibration selections prefer MSP over their best normalized candidate. Test comparisons below nevertheless report normalization itself as a sensitivity, without selecting a score from the test results. Predictions and full-coverage risk do not change. No probability required clipping to compute its logarithm.

Table S21: Calibration-selected normalization exponents. The fallback to MSP is the same for every row and seed.

| Backbone | State | Target | 17 | 42 | 89 |
|---|---|---|---|---|---|
| Swin-T | Selected | Hard | 2 | 2 | 3 |
| Swin-T | Selected | Tie-hard | 1.5 | 2 | 2 |
| Swin-T | Selected | Uniform | 1.5 | 3 | 1.5 |
| Swin-T | Selected | Tie-Uniform | 2 | 3 | 4 |
| Swin-T | Selected | Soft | 4 | 4 | 3 |
| Swin-T | Epoch 12 | Hard | 1.5 | 1.5 | 1.5 |
| Swin-T | Epoch 12 | Tie-hard | 1 | 1.5 | 1.5 |
| Swin-T | Epoch 12 | Uniform | 1.5 | 1.5 | 1.5 |
| Swin-T | Epoch 12 | Tie-Uniform | 2 | 2 | 2 |
| Swin-T | Epoch 12 | Soft | 4 | 4 | 3 |
| ResNet-18 | Selected | Soft | 3 | 3 | 3 |
| ResNet-18 | Selected | Uniform | 3 | 1.5 | 8 |
| ResNet-18 | Selected | Tie-Uniform | 2 | 1.5 | 2 |
| ResNet-18 | Epoch 12 | Soft | 2 | 3 | 3 |
| ResNet-18 | Epoch 12 | Uniform | 3 | 3 | 3 |
| ResNet-18 | Epoch 12 | Tie-Uniform | 2 | 2 | 2 |

Table S22: Soft-minus-control disagreement at 80% coverage (percentage points). Each model and score retains its own 2,620 images. Seed columns are paired fitted comparisons, not confidence limits.

| Backbone | State | Control | Score | 17 | 42 | 89 | Mean |
|---|---|---|---|---|---|---|---|
| Swin-T | Selected | Uniform | MSP | -0.347 | -0.431 | -0.355 | -0.378 |
| Swin-T | Selected | Uniform | Norm. | -0.141 | -0.305 | -0.221 | -0.223 |
| Swin-T | Selected | Tie-Uniform | MSP | -0.378 | -0.584 | -0.802 | -0.588 |
| Swin-T | Selected | Tie-Uniform | Norm. | -0.034 | -0.176 | -0.794 | -0.335 |
| Swin-T | Epoch 12 | Uniform | MSP | -0.328 | -0.393 | -0.229 | -0.317 |
| Swin-T | Epoch 12 | Uniform | Norm. | -0.088 | -0.122 | -0.156 | -0.122 |
| Swin-T | Epoch 12 | Tie-Uniform | MSP | -0.435 | -0.221 | -0.263 | -0.307 |
| Swin-T | Epoch 12 | Tie-Uniform | Norm. | -0.172 | -0.176 | -0.042 | -0.130 |
| ResNet-18 | Selected | Uniform | MSP | +0.347 | +0.137 | +0.034 | +0.173 |
| ResNet-18 | Selected | Uniform | Norm. | +0.435 | +0.324 | +0.305 | +0.355 |
| ResNet-18 | Selected | Tie-Uniform | MSP | +0.118 | +0.115 | -0.107 | +0.042 |
| ResNet-18 | Selected | Tie-Uniform | Norm. | -0.134 | +0.370 | +0.229 | +0.155 |
| ResNet-18 | Epoch 12 | Uniform | MSP | +0.073 | +0.489 | -0.050 | +0.170 |
| ResNet-18 | Epoch 12 | Uniform | Norm. | +0.313 | +0.718 | +0.347 | +0.459 |
| ResNet-18 | Epoch 12 | Tie-Uniform | MSP | +0.324 | +0.118 | +0.126 | +0.190 |
| ResNet-18 | Epoch 12 | Tie-Uniform | Norm. | +0.492 | +0.481 | +0.389 | +0.454 |

## S10.2 Target entropy and weighted class proportions

Entropy is computed on positive supported-mass Training rows before fitting. The weighted mean uses the same supported masses as the training loss. Class-marginal L1 distance compares the weighted target

distribution with Soft within each vocabulary. Zero-mass rows have zero target loss and are excluded from unweighted entropy means. The permutation control exactly preserves each seven-output probability multiset and all maximum positions; it does not preserve aggregate class proportions. These descriptive checks do not identify a causal mechanism.

Table S23: Training-target properties before outcome evaluation. Entropy is in nats; all conditions have 28,709 Training images.

| Outputs | Target | Positive rows | Entropy | Weighted | L1 |
|---|---|---|---|---|---|
| 7 | Soft | 28557 | 0.5067 | 0.4949 | 0.0000 |
| 7 | Uniform | 28557 | 0.8002 | 0.7812 | 0.1197 |
| 7 | Tie-Uniform | 28557 | 0.7727 | 0.7553 | 0.1032 |
| 7 | Permutation | 28557 | 0.5067 | 0.4949 | 0.1242 |
| 8 | Soft | 28559 | 0.5399 | 0.5298 | 0.0000 |
| 8 | Uniform | 28559 | 0.8665 | 0.8505 | 0.1295 |
| 8 | Tie-Uniform | 28559 | 0.8381 | 0.8231 | 0.1111 |

## S10.3 Fixed middle-window aggregation

The middle four chronologically sampled positions (2–5, zero-based) are averaged without selecting peak frames. Each aggregation reranks clips independently by MSP. All 1,440 clips have eight valid retained frames; the original eight-frame aggregation reproduces every stored clip probability exactly. Human votes describe whole clips rather than individual frames. The middle-window analysis cannot determine which frames express the intended or perceived category.

Table S24: External Soft-minus-Uniform disagreement at 80% coverage (percentage points). Each state and aggregation accepts 1,152 clips under its own ranking.

| Backbone | State | Votes | Frames | 17 | 42 | 89 | Mean |
|---|---|---|---|---|---|---|---|
| Swin-T | Selected | Raw | Eight | -0.05 | +2.64 | -0.76 | +0.61 |
| Swin-T | Selected | Raw | Middle four | +1.00 | +1.70 | -0.92 | +0.59 |
| Swin-T | Selected | Merged | Eight | -1.28 | +2.48 | -0.90 | +0.10 |
| Swin-T | Selected | Merged | Middle four | -0.22 | +0.43 | -0.83 | -0.21 |
| Swin-T | Epoch 12 | Raw | Eight | -0.46 | +1.02 | +0.11 | +0.23 |
| Swin-T | Epoch 12 | Raw | Middle four | +1.55 | +0.54 | -0.21 | +0.63 |
| Swin-T | Epoch 12 | Merged | Eight | -1.70 | -0.33 | -0.68 | -0.90 |
| Swin-T | Epoch 12 | Merged | Middle four | +0.46 | -0.43 | -0.77 | -0.25 |
| ResNet-18 | Selected | Raw | Eight | +1.04 | +5.52 | -1.07 | +1.83 |
| ResNet-18 | Selected | Raw | Middle four | +0.24 | +5.73 | -0.25 | +1.91 |
| ResNet-18 | Selected | Merged | Eight | +1.11 | +3.78 | -0.91 | +1.33 |
| ResNet-18 | Selected | Merged | Middle four | -0.14 | +3.21 | -0.10 | +0.99 |
| ResNet-18 | Epoch 12 | Raw | Eight | +2.20 | +3.06 | -0.15 | +1.70 |
| ResNet-18 | Epoch 12 | Raw | Middle four | +3.12 | +5.10 | +1.16 | +3.12 |
| ResNet-18 | Epoch 12 | Merged | Eight | +2.27 | +1.81 | -0.23 | +1.28 |
| ResNet-18 | Epoch 12 | Merged | Middle four | +3.24 | +3.63 | +0.99 | +2.62 |

The full ledger retains all 36 frozen external states, both reference mappings, all-eligible and 80% populations, risk components and category proportions. Constant-category references have no confidence ranking. They are scored both on all clips and on each model's exact accepted set, so the two population types are not interchangeable. Always-neutral disagreement on all clips is 90.94% raw and 82.18% calm-merged; always-happy disagreement is 83.23% for either reference. These references establish poor absolute transfer despite performance better than these constant predictions.

# S11 New target, vocabulary and stopping-rule comparisons

The final protocol defines twelve additional Swin-T runs on the same previously exposed source. Seven-output Permutation uses the three existing seeds; eight-output Soft, Uniform and Tie-Uniform use the same three seeds and budget. All twelve runs were completed and hashed before new calibration/test inference. There are 33 nominal inference states: six for Permutation and 27 for the eight-output conditions. Different policies can select the same epoch and therefore do not necessarily represent distinct weights. These comparisons remain adaptive within-source evidence.

All risks retain ten annotation categories. Each model selects its own 2,620 primary-test images at 80% MSP coverage. Squared distance uses all 3,275 images. Intervals below use 5,000 shared exact-pixel-cluster replicates, rerank each resampled multiset, and average the three seed contrasts within each replicate. They condition on fitted models and observed annotations, rather than estimating variation over future seeds or people.

Table S25: Every new fitted state: complete-vote disagreement at 80% coverage (%) and full-coverage ten-category squared distance.

| Outputs | Target | State | Seed | Risk | Distance |
|---|---|---|---|---|---|
| 7 | Permutation | Distance-selected | 17 | 24.225 | 0.12121 |
| 7 | Permutation | Epoch 12 | 17 | 24.126 | 0.12496 |
| 7 | Permutation | Distance-selected | 42 | 24.676 | 0.12280 |
| 7 | Permutation | Epoch 12 | 42 | 24.248 | 0.12511 |
| 7 | Permutation | Distance-selected | 89 | 24.534 | 0.12077 |
| 7 | Permutation | Epoch 12 | 89 | 24.168 | 0.12400 |
| 8 | Soft | Distance-selected | 17 | 23.767 | 0.09472 |
| 8 | Soft | Epoch 12 | 17 | 23.492 | 0.09417 |
| 8 | Soft | Risk-selected | 17 | 23.492 | 0.09417 |
| 8 | Uniform | Distance-selected | 17 | 24.221 | 0.11632 |
| 8 | Uniform | Epoch 12 | 17 | 23.893 | 0.11533 |
| 8 | Uniform | Risk-selected | 17 | 24.221 | 0.11632 |
| 8 | Tie-Uniform | Distance-selected | 17 | 23.779 | 0.10877 |
| 8 | Tie-Uniform | Epoch 12 | 17 | 23.882 | 0.11000 |
| 8 | Tie-Uniform | Risk-selected | 17 | 23.882 | 0.11000 |
| 8 | Soft | Distance-selected | 42 | 23.565 | 0.09426 |
| 8 | Soft | Epoch 12 | 42 | 23.565 | 0.09426 |
| 8 | Soft | Risk-selected | 42 | 23.786 | 0.09521 |
| 8 | Uniform | Distance-selected | 42 | 24.756 | 0.11772 |
| 8 | Uniform | Epoch 12 | 42 | 24.015 | 0.11603 |
| 8 | Uniform | Risk-selected | 42 | 23.950 | 0.11561 |
| 8 | Tie-Uniform | Distance-selected | 42 | 23.615 | 0.10881 |
| 8 | Tie-Uniform | Epoch 12 | 42 | 23.615 | 0.10881 |
| 8 | Tie-Uniform | Risk-selected | 42 | 23.615 | 0.10881 |
| 8 | Soft | Distance-selected | 89 | 23.893 | 0.09503 |
| 8 | Soft | Epoch 12 | 89 | 23.828 | 0.09432 |
| 8 | Soft | Risk-selected | 89 | 23.893 | 0.09503 |
| 8 | Uniform | Distance-selected | 89 | 24.756 | 0.11719 |
| 8 | Uniform | Epoch 12 | 89 | 24.134 | 0.11680 |
| 8 | Uniform | Risk-selected | 89 | 24.756 | 0.11719 |
| 8 | Tie-Uniform | Distance-selected | 89 | 23.989 | 0.11129 |
| 8 | Tie-Uniform | Epoch 12 | 89 | 23.779 | 0.11032 |
| 8 | Tie-Uniform | Risk-selected | 89 | 23.989 | 0.11129 |

Table S26: Soft-minus-control seed differences, means, sample standard deviations and conditional 95% intervals (all percentage points). Negative favors Soft. Distance and Risk denote the selection objectives.

| K | Control | State | 17 | 42 | 89 | Mean | SD | Interval |
|---|---|---|---|---|---|---|---|---|
| 7 | Permutation | Distance | -0.523 | -0.935 | -0.924 | -0.794 | 0.235 | [-1.071, -0.474] |
| 7 | Permutation | Epoch 12 | -0.481 | -0.557 | -0.607 | -0.548 | 0.063 | [-0.855, -0.261] |
| 8 | Uniform | Distance | -0.454 | -1.191 | -0.863 | -0.836 | 0.369 | [-1.152, -0.509] |
| 8 | Tie-Uniform | Distance | -0.011 | -0.050 | -0.095 | -0.052 | 0.042 | [-0.314, +0.212] |
| 8 | Uniform | Epoch 12 | -0.401 | -0.450 | -0.305 | -0.385 | 0.074 | [-0.672, -0.071] |
| 8 | Tie-Uniform | Epoch 12 | -0.389 | -0.050 | +0.050 | -0.130 | 0.230 | [-0.388, +0.141] |
| 8 | Uniform | Risk | -0.729 | -0.164 | -0.863 | -0.585 | 0.371 | [-0.892, -0.282] |
| 8 | Tie-Uniform | Risk | -0.389 | +0.172 | -0.095 | -0.104 | 0.281 | [-0.357, +0.168] |

## S11.1 Fixed-image cross-evaluation

The following averages apply each predictor to both selectors' accepted images. They distinguish prediction changes on fixed images from the changes in image composition inherent in own-ranking comparisons. They are descriptive cross-evaluations, not causal decomposition.

Table S27: Fixed-set risks (%). S is Soft and C is the named control; predictor appears before selector in each heading. All cells retain 2,620 identical images within a selector and seed.

| K | Control | State | S on S | C on S | S on C | C on C |
|---|---|---|---|---|---|---|
| 7 | Permutation | Distance-selected | 23.684 | 24.207 | 24.405 | 24.478 |
| 7 | Permutation | Epoch 12 | 23.632 | 24.048 | 24.092 | 24.181 |
| 8 | Uniform | Distance-selected | 23.742 | 24.471 | 24.445 | 24.578 |
| 8 | Tie-Uniform | Distance-selected | 23.742 | 24.020 | 23.919 | 23.794 |
| 8 | Uniform | Epoch 12 | 23.628 | 24.121 | 24.083 | 24.014 |
| 8 | Tie-Uniform | Epoch 12 | 23.628 | 23.868 | 23.860 | 23.758 |
| 8 | Uniform | Risk-selected | 23.724 | 24.314 | 24.332 | 24.309 |
| 8 | Tie-Uniform | Risk-selected | 23.724 | 24.003 | 23.955 | 23.828 |

## S11.2 Selection objectives and supported-mass strata

Distance selection minimizes ten-category squared distance; the risk rule minimizes complete-vote disagreement at 80% MSP coverage on the selection subset. Both retain the earliest minimizing epoch. The risk rule is reported only for the eight-output family and is not retrospectively reconstructed for the original runs.

Table S28: Epochs selected under the declared policies. A dash indicates a policy not reported for that family.

| Outputs | Target | Seed | Distance | Risk | Fixed |
|---|---|---|---|---|---|
| 7 | Permutation | 17 | 9 | – | 12 |
| 7 | Permutation | 42 | 7 | – | 12 |
| 7 | Permutation | 89 | 7 | – | 12 |
| 8 | Soft | 17 | 9 | 12 | 12 |
| 8 | Soft | 42 | 12 | 9 | 12 |
| 8 | Soft | 89 | 10 | 10 | 12 |
| 8 | Uniform | 17 | 9 | 9 | 12 |
| 8 | Uniform | 42 | 6 | 11 | 12 |
| 8 | Uniform | 89 | 6 | 6 | 12 |
| 8 | Tie-Uniform | 17 | 9 | 12 | 12 |
| 8 | Tie-Uniform | 42 | 12 | 12 | 12 |
| 8 | Tie-Uniform | 89 | 10 | 10 | 12 |

Table S29: Eight-output distance-selected supported-mass strata: means over seeds. Risk is a percentage; components are percentage-point contributions. Accepted counts can vary by seed. The full CSV retains every state and scope.

| Target | Stratum | Scope | N | Kept | Risk | Unsupp. | Excess |
|---|---|---|---|---|---|---|---|
| Soft | full | all | 1873 | 1873.0 | 19.55 | 0.00 | 3.22 |
| Soft | full | global80 | 1873 | 1639.0 | 14.64 | 0.00 | 1.48 |
| Soft | full | within80 | 1873 | 1499.0 | 12.18 | 0.00 | 0.90 |
| Soft | partial | all | 1386 | 1386.0 | 45.78 | 13.79 | 6.51 |
| Soft | partial | global80 | 1386 | 969.3 | 38.22 | 13.21 | 3.30 |
| Soft | partial | within80 | 1386 | 1109.0 | 40.64 | 13.43 | 4.32 |
| Soft | zero | all | 16 | 16.0 | 100.00 | 100.00 | 0.00 |
| Soft | zero | global80 | 16 | 11.7 | 100.00 | 100.00 | 0.00 |
| Soft | zero | within80 | 16 | 13.0 | 100.00 | 100.00 | 0.00 |
| Uniform | full | all | 1873 | 1873.0 | 20.46 | 0.00 | 4.14 |
| Uniform | full | global80 | 1873 | 1628.0 | 15.14 | 0.00 | 1.86 |
| Uniform | full | within80 | 1873 | 1499.0 | 12.76 | 0.00 | 1.15 |
| Uniform | partial | all | 1386 | 1386.0 | 46.51 | 13.79 | 7.24 |
| Uniform | partial | global80 | 1386 | 983.3 | 39.54 | 13.18 | 4.09 |
| Uniform | partial | within80 | 1386 | 1109.0 | 41.41 | 13.35 | 4.99 |
| Uniform | zero | all | 16 | 16.0 | 100.00 | 100.00 | 0.00 |
| Uniform | zero | global80 | 16 | 8.7 | 100.00 | 100.00 | 0.00 |
| Uniform | zero | within80 | 16 | 13.0 | 100.00 | 100.00 | 0.00 |
| Tie-Uniform | full | all | 1873 | 1873.0 | 19.73 | 0.00 | 3.40 |
| Tie-Uniform | full | global80 | 1873 | 1639.7 | 14.75 | 0.00 | 1.45 |
| Tie-Uniform | full | within80 | 1873 | 1499.0 | 12.17 | 0.00 | 0.84 |
| Tie-Uniform | partial | all | 1386 | 1386.0 | 45.95 | 13.79 | 6.68 |
| Tie-Uniform | partial | global80 | 1386 | 973.0 | 38.45 | 13.15 | 3.38 |
| Tie-Uniform | partial | within80 | 1386 | 1109.0 | 40.81 | 13.38 | 4.29 |
| Tie-Uniform | zero | all | 16 | 16.0 | 100.00 | 100.00 | 0.00 |
| Tie-Uniform | zero | global80 | 16 | 7.3 | 100.00 | 100.00 | 0.00 |
| Tie-Uniform | zero | within80 | 16 | 13.0 | 100.00 | 100.00 | 0.00 |

Full and partial denote supported vote mass one and strictly between zero and one; zero denotes no supported votes. Global 80% restricts the full-population accepted set; within-stratum 80% reranks within the stratum. Supported ambiguity is risk minus unsupported mass minus excess before rounding. A dash in a mean denotes an undefined value for at least one seed, rather than averaging only nonempty seeds. Empty or unsupported-only conditional risks are not replaced with zero. Every 20%, 50%, 80% and 100% summary, all risk–coverage prefixes and fixed-image per-seed cells are retained in the result ledgers.

## S11.3 Fixed confidence-ranking extension

The extension covers 33 new states and six historical Soft comparators: 81 named states together with the original analysis, not 87 independent fits. It was frozen during training before new calibration/test inference, after earlier outcomes were known. Calibration preferred MSP for every new state; no zero clipping was needed. Ledgers retain all seeds, coverages and candidate exponents. Table S30 is descriptive, without new bootstrap intervals.

Table S30: Fixed ranking sensitivities for all new paired comparisons at 80% coverage. Values are means of three paired seed differences, in percentage points; negative favors Soft. Norm. uses a calibration-selected exponent, and Choice compares only MSP with Norm. on calibration AURC. These are adaptive descriptive sensitivities, not new hypothesis tests.

| K | Control | State | MSP | Margin | Neg. entropy | Norm. | Choice |
|---|---|---|---|---|---|---|---|
| 7 | Permutation | Distance | -0.794 | -0.546 | -0.604 | -0.663 | -0.794 |
| 7 | Permutation | Epoch 12 | -0.548 | -0.244 | -0.630 | -0.341 | -0.548 |
| 8 | Uniform | Distance | -0.836 | -0.422 | -0.588 | -0.457 | -0.836 |
| 8 | Tie-Uniform | Distance | -0.052 | +0.280 | +0.164 | +0.131 | -0.052 |
| 8 | Uniform | Epoch 12 | -0.385 | +0.003 | -0.089 | -0.117 | -0.385 |
| 8 | Tie-Uniform | Epoch 12 | -0.130 | +0.183 | +0.179 | +0.101 | -0.130 |
| 8 | Uniform | Risk | -0.585 | -0.242 | -0.253 | -0.417 | -0.585 |
| 8 | Tie-Uniform | Risk | -0.104 | +0.258 | +0.154 | +0.109 | -0.104 |